SAAB AM – Fluid and heat analysis

Liquid cooler design for electronics in future aircrafts using powder bed additive manufacturing

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Abstract
This report explains how and why the heat and fluid analysis have been conducted as well as the practical experiments that have been conducted in the work process. The purpose of the analyses is to investigate two concepts for a liquid cooler that is supposed to cool a specified area of a PC box. The cooling liquid is not allowed to increase by more than 5 degrees centigrade and shall disperse of 500W of energy.

The analyses have been apart of the design exploration conducted in this project, read more about the design exploration in the sub-report "*Design Space Exploration and Optimization*". The design exploration uses a simplification of the Ansys CFX model where the entire model is simplified as only one channel to save computational time and to allow a more flexible variation in the parameters. Since Ansys cannot adapt meshes if the number of surfaces changes in the model. To be able to perform the analyses and verify the flow division design, a number of experiments had to be performed before the CFX analyses are performed. The results from the design exploration are parameters for the theoretically best design, which will then be used to analyze the full model.

The results from the analyses shows that both concepts are able to meet the goal of dispersing the 500W of energy while not increasing the cooling liquid by more than 5 degrees centigrade. The analyses in the design exploration do not consider fluid flow and assumes that the flow is divided evenly across all channels. This is not the case and is proven by the full model analysis, the fluid flow division varies between the two concepts where one is clearly better than the other.

The most critical problem areas brought up in this report are due to the simplifications and assumptions made to be able to finish, thus it is strongly recommended that the designs are further investigated by a more extensive heat transfer analysis as well as including fluid flow in the design evaluation and exploration.

Nomenclature

The following section defines acronyms and terms that have been used in this report.

Nu	Nusselt number	[-]
Pr	Prandtl number	[-]
Re_d	Reynolds number	[-]
L_c	Characteristic length	[m]
h	Heat transfer coefficient	[W/m^2-K]
k	Thermal conductivity	[W/m-K]
υ	Kinematic viscosity	[m^2/s]
d_h	Hydraulic diameter	[m]
A	Cross-sectional area	[m^2]
O	Circumference	[m]
μ	Dynamic viscosity	[N s/m^2]
cp	Specific heat	[J/kg-K]
U	Flow velocity	[m/s]
ς	Energy loss factor	[-]
λ	Darcy friction factor	[-]
ρ	Density	[kg/m^3]
$\bar{u}$	Mean flow velocity	[m/s]
L	Pipe or channel length	[m]
d	Pipe or channel diameter	[m]
r	Pipe or channel radius	[m]

Definitions

Substrate – The solid material part of the cooler, i.e. the actual cooler that the fluid runs through

Design exploration – A simplified optimization that serves to explore the design space and point to a general area where an optimum might be

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1. Introduction
SAAB aeronautic design and produces airplanes and airplane systems both for commercial and military purpose. Today's costumers demand shorter development times and better products. By reducing time for construction, production and assembly this could be achieved. By using additive manufacturing (AM) there is a potential to reduce the time from idea to final product but also to design geometries that are not possible with convention manufacturing methods. More complex geometries have the potential of increasing the functionality and reduce the weight of a component.

With computer power ever increasing it puts a higher pressure on cooling of the electronic components. In today's aircrafts air cooling is enough but in the future, 2040 and onwards, a larger need for cooling may be required. The user demands more out of less which means a higher effect on smaller surfaces. At the same time weight needs to be minimized which creates a difficult problem to solve. A ground demonstrator is set to be finished in the middle of the 2020's which means the cooling issue needs to be solved today before the demonstrator is constructed.

1.1. Purpose of this report
The objective of the heat- and flow analysis is to determine the temperature difference between the inlet and outlet for the cooler as well as see that the flow distributes evenly through the cooling channels to create even cooling.

1.2. Delimitations
The information provided by SAAB was to assume an inlet pump pressure of 8 bar, due to lack of access to a pump matching that criteria, a pump with an output pressure of 4.8 bar was used in the experiments.

The Ansys CFX simulations are done assuming steady state at operating time, due to the nature of the simulation mentioned earlier under chapter 1, the pc box and substrate are not included. Thus 100% of the energy emitted from the pc box goes into the substrate and is then distributed evenly across the entire channel surface.

2. Flow testing

To verify the concepts that passed the screening and scoring (concept 3 and concept 5, see the main report under chapter 5), flow testing was performed. The objective was to determine if the design was able to distribute flow evenly through the channels. A traditional channel split is made from one into two channels, but due to the construction volume being defined within the cooling area the amount of space for channel division is limited. Since the cooling is desired to be even across the entire surface, another design had to be evaluated (see Figure 1). The traditional channel split guarantees even flow and pressure division, the design evaluated is reminiscent of a manifold.

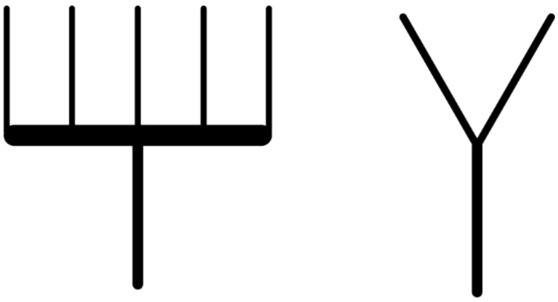

Figure 1 - Left: Manifold design evaluated   Right: Traditional channel split

Another part of the test is to see if the connection to the outlet works in a good way. For example if it is possible to collect the water without using too much of the construction volume. As well as getting flow rate data for the heat transfer and flow simulations.

2.1. Test plan

The specified inlet pressure was 8 bar but the pump used had an output pressure of 4,8 bar with an outlet diameter of 25,4 mm or 1 inch. This means that the inlet outside diameter to the model had to be the same for the tubes to fit, which in turns means that the inlet to the models had to have a choke from 25,4 mm to 15 mm in order to obtain a sufficient wall thickness.

There was a total of six tests that were performed:

1. Measure flow from the pump
2. Measure flow after the inlet choke
3. Test half model of concept 3 with 29 channels and measure flow through five channels
4. Test full model of concept 3 and measure outlet flow
5. Test full model of concept 5 with 14 channels and see if flow is distributed evenly
6. Test altered, full model of concept 5 with center rib and 16 channels

The test models can be viewed under
Appendix I – 3D printed test models.

To measure the flow rate a rather crude method was used, an empty container was weighed before the testing and then filled with water during the tests and weighed after to determine the amount of water. It was deemed a better method than measuring, the water density was set at 20 degrees centigrade (not measured but assumed temperature).

Test equipment:

- Pump, Jet pump BP 801 from Biltema
    - Pressure 4,8 bar
    - Outlet diameter 25,4 mm (1 inch)
- Container to collect water
- Reservoir

- Test models
    - Concept 3, half model
    - Concept 3, full model
    - Concept 5, full model
    - Concept 5, full modified model
    - Inlet choke
- Screw driver
- Wrench
- Pliers
- 3x hose clamps
- 2x hose couplings
- Sealing tape

2.2. Test results

The results from each test are presented in this chapter.

2.2.1. Test 1 – Outlet flow rate from pump

This test was performed to measure the flow rate from the pump, directly from the 25,4 mm tube connected to it. In total two measures were performed to create get a mean value for the flow and to see that the tests yield similar results.

Table 1 - Pump outlet flow rate measures

Measure [Nr]	Total weight [g]	Weight of water [g]	Time [s]	Mass flow rate [kg/s]	Volumetric flow rate [m^3/s]	Flow velocity [m/s]
1	11391	9801	10	0,9801	0,0009819	1,94
2	11503	9913	10	0,9913	0,0009931	1,96

Table 2 - Mean outlet flow rates calculated

Mean mass flow rate [kg/s]	Mean volumetric flow rate [m^3/s]	Mean flow velocity [m/s]
0,9857	0,0009875	1,95

2.2.2. Test 2 – Flow rate through inlet choke

This test has the inlet choke connected to the pump outlet tube, to measure the mass flow rate going in to the concepts. Which is needed for the simulations, only one measure was performed for this test.

Table 3 - Inlet choke flow rate measure

Measure [Nr]	Total weight [g]	Weight of water [g]	Time [s]	Mass flow rate [kg/s]	Volumetric flow rate [m^3/s]	Flow velocity [m/s]
1	10961	9371	10	0,9371	0,0009388	5,31

2.2.3. Test 3 – Concept 3, half model

This test was performed on a half model of concept 3 to measure the flow through the channels to determine whether or not the chamber design is able to divide flow evenly through the model. The model had 29 channels, but in order to physically be able to fit tubing on the outlets only five channels were fitted with nippels, evenly spread out across the channels. Illustrated in Figure 2, Figure 3 shows the actual test being conducted. The results from the tests are available in Table 4, and calculated mean values are available in Table 5. This model can be seen under Appendix I – 3D printed test models, Figure 30.

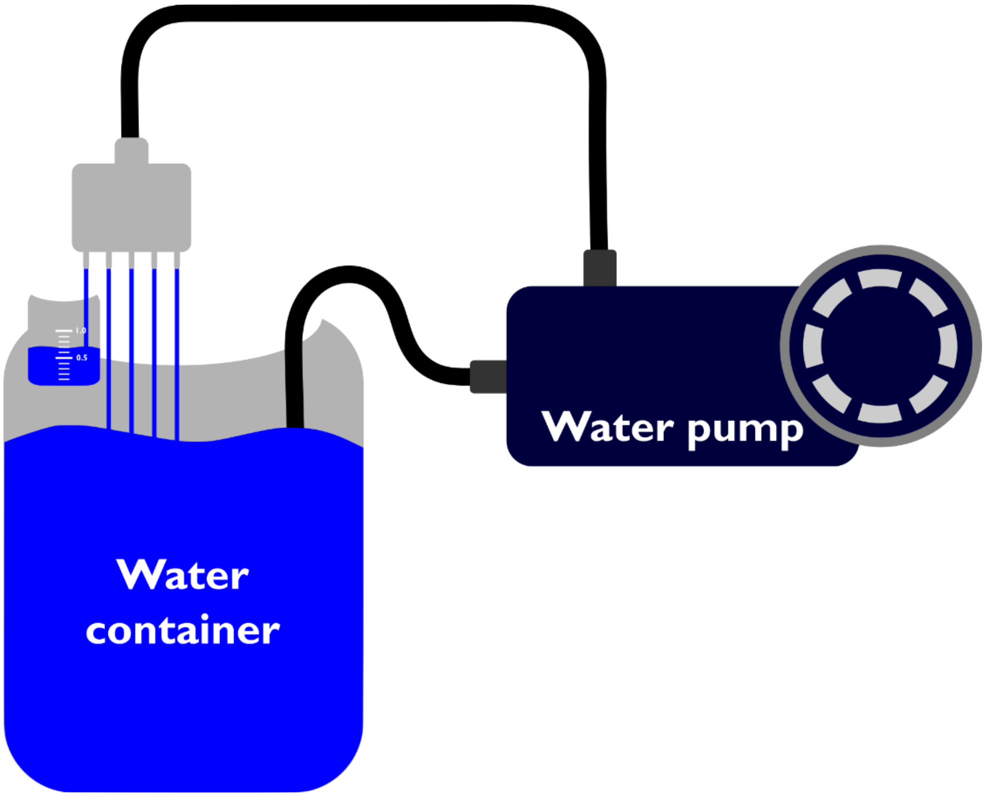

Figure 2 - Schematic picture of test 3

Figure 3 - Picture from test 3

Table 4 - Outlet flow rates from test 3

Outlet Position	Measure [Nr]	Total weight [g]	Weight of water [g]	Time [s]	Mass flow rate [kg/s]	Volumetric flow rate [m^3/s]	Flow velocity [m/s]
Left	1	898	739	30	0,02463	0,00002468	3,49
Left	2	899	740	30	0,02467	0,00002471	3,50
Left	3	906	747	30	0,02490	0,00002494	3,53
Left mid	1	880	721	30	0,02403	0,00002408	3,41
Left mid	2	882	723	30	0,02410	0,00002414	3,42
Left mid	3	875	716	30	0,02387	0,00002391	3,38
Middle	1	1183	1024	25	0,04096	0,00004103	5,81
Middle	2	1214	1055	25	0,04220	0,00004228	5,98
Middle	3	1203	1044	25	0,04176	0,00004184	5,92
Right mid	1	879	720	30	0,02400	0,00002404	3,40
Right mid	2	886	727	30	0,02423	0,00002428	3,43
Right mid	3	883	724	30	0,02413	0,00002148	3,42
Right	1	950	791	30	0,02637	0,00002641	3,74
Right	2	963	804	30	0,02680	0,00002685	3,80
Right	3	974	815	30	0,02717	0,00002722	3,85

To get a more accurate result three measures were conducted on each outlet to obtain a mean value.

Table 5 - Mean values of outlet flow rates calculated

Outlet position	Mean mass flow rate [kg/s]	Mean volumetric rate [m^3/s]	Mean flow velocity [m/s]
Left	0,02473	0,00002478	3,51
Left mid	0,02400	0,00002404	3,40
Middle	0,04164	0,00004172	5,90
Right mid	0,02412	0,00002417	3,42
Right	0,02678	0,00002683	3,80

Something that can be observed from these results directly is that there is a slight difference in flow rates between the left and the right channel even though the model has completely symmetric sides. This anomaly may be due to leakage between left outlet nipple and the tube connected, which was observed during the testing on the left side but not on the right side.

2.2.4. Test 4 – Concept 3, full model

This test was performed on the entire model of concept 3 to measure the outlet flow. This model can be seen under
Appendix I – 3D printed test models, Figure 31.

Table 6 - Concept 3 outlet flow rate measures

Measure [Nr]	Total weight [g]	Weight of water [g]	Time [s]	Mass flow rate [kg/s]	Volumetric flow rate [m^3/s]	Flow velocity
1	10948	9517	10	0,9517	0,0009534	5,40
2	10228	8956	10	0,8956	0,0008972	5,08

Table 7 - Concept 3 mean value flow rates calculated

Mean mass flow rate [kg/s]	Mean volumetric flow rate [m^3/s]	Mean flow velocity [m/s]
0,9237	0,0009253	5,24

2.2.5. Test 5 – Concept 5, full model

This test was performed to verify that the flow is distributed evenly through concept 5 as well. Since the channels are square it was not possible to make the same measurements as for concept 3, instead this was made with a see-through bottom plate to visually observe the flow. Hence, only the outlet flow was measured. This model can be seen under
Appendix I – 3D printed test models, Figure 32.

Two measures were performed to measure the outlet flow.

Table 8 - Concept 5, full model flow rate measures

Measure [Nr]	Total Weight [g]	Weight of water [g]	Time [s]	Mass flow rate [kg/s]	Volumetric flow rate [m^3/s]	Flow velocity
1	10604	9173	10	0,9173	0,0009190	5,20
2	10812	9381	10	0,9381	0,0009398	5,32

Table 9 - Concept 5, full model mean flow rates calculated

Mean mass flow rate [kg/s]	Mean volumetric flow rate [m^3/s]	Mean flow velocity [m/s]
0,9277	0,0009294	5,26

2.2.6. Test 6 – Concept 5, full modified model

Since the flow was not evenly distributed through the channels for concept 5, another model was created with slight changes in design to help the flow distribute better. Since there was no way of measuring the flow through the channels because they are square, a simple visual inspection was performed. The flow distribution will be further investigated using CFD-models later in the project.

2.2.7. Error sources

Due to the nature of the experiments, being conducted with lack of access to proper equipment there are several error sources present. They will be presented in this chapter but the impact will be discussed in chapter 6, Discussion.

During testing of the pump and during the tests there were leaks present between the hose couplings connecting the tubes to the pump, seen in Figure 4.

Figure 4 - Hose couplings connceting tubes to the pump

In order to measure flow through the channels in test 3, some channels were fitted with outlet nippels to be able to connect tubes and divert the flow to a separate container. These nippels and the tubes attached will affect the flow through these channels since they add “resistance” to the channels and flow favors the path of least resistance.

The tubes connected to the pump became somewhat deformed during the testing, this means that creases appeared at some places on the tubes, this can be also be seen in Figure 4, at the very top.

We are using a coupling which changes the inlet diameter from 25,4 mm to 15 mm which will affect the inlet pressure, which means we will not have exactly 4.8 bar.

Since the test models are 3D-printed using PLA plastic there may be some leakage within the model that is not visible and thus can affect the flow.

The outlet connections appeared to be leaking during testing, the left outlet also deformed during the fitting of the tube since the tubes had to heated to fit. The left nipple also appeared to have a greater visible leak than the other outlets that did not display any visible leakage.

The tests were timed manually and because of it the reaction time of starting and stopping the timer and the flow may have affected the results.

3. Heat and flow analysis

The first chapter explores the impact of the fluid temperature increase on the fluid parameters since only the inlet temperature is known when the heat and flow analyses are performed. The second and third chapter contains calculations and simulations specifically for the two chosen concepts (for more information about the concept development see the project report under chapter 3).

Since Ansys cannot adapt the mesh automatically to a change in the number of faces of the model it creates a problem with the design optimization. There is also a problem with computational time for the Ansys models in the optimization, ModeFrontier triggers the project update function in the Ansys benchwork which takes longer time than running it manually. Thus, if the entire model was to be analyzed the time it would take to run all DOE's would be to long within the given timeframe. To compensate for these issues, a design exploration is performed to indicate in which area the optimal solution lies. The exploration is not a full optimization, more on this subject can be found in the sub-report "*Design Space Exploration and Optimization*". This means that two different frameworks for each concept hade to be created in Ansys, one for the one-channel approximation and one for the full model.

The analyses disregard the solid material all together, since the heat transfer analysis would need the fluid bulk mean temperature and heat transfer coefficient it creates problems; the purpose of these analyses is to find the fluid temperature and the heat transfer coefficient would differ for each channel. Which is possible to calculate but the fluid velocities are unknown.

Instead a simpler fluid analysis is made, where some assumptions are made to make it work. First, an assumption is made that 100% of the energy emitted from the PC box is absorbed by the cooler substrate; second, the heat absorbed by the cooler substrate is distributed evenly across all of the channel surface.

3.1. Fluid temperature impact on fluid properties

According to (Rohsenow, Hartnett, & Cho, 1998), the most useful parameters in heat transfer are the heat transfer coefficient and the fluid bulk mean temperature. Normally the bulk mean fluid temperature would be used for the simulations but since the objective is to determine the fluid temperature increase in the cooler this causes a problem. Since the inlet temperature is known, coupled with the fact that the cooling problem is very simple, the impact of the temperature increase within the 5 degrees centigrade interval on the fluid properties is investigated.

Calculations are made for the test model parameters, which is concept 3 (see Appendix I – 3D printed test models, Figure 31). The test model had 29 channels with a 3 mm channel diameter and an inlet diameter of 15 mm. The channel length is 275 mm, the total cooler length is 320 mm and 125 mm wide. Since the channels make up almost 86% of the total length

The fluid is Dowcal100 with 50% ethylene glycol concentration at an inlet temperature of 70 degrees centigrade. See Appendix II – Dowcal100 property tables for Dowcal100 properties, however the properties are not given for 70 degrees centigrade, this is examined further in this chapter. The fluid temperature cannot increase by more than 5 degrees centigrade between the inlet and outlet. The energy addition is 500W to the bottom of the cooler substrate.

Note; from this point on Dowcal100 with 50% ethylene glycol concentration will simply be referred to as Dowcal100.

To get the fluid properties of Dowcal100 a linear interpolation is made which gave the following values:

Table 10 - Fluid properties of Dowcal100 at 70 degrees centigrade

Fluid property	Value	Unit
Specific heat	3516	J/kg-K
Density	1039,8	kg/m^3
Thermal conductivity	0,4148	W/m-K
Dynamic viscosity	0,001414	kg/ms
Kinematic viscosity	1,34452e-06	m^2/s
Thermal diffusivity	1,13459e-07	m^2/s

Since the channels make up almost 86% of the total cooler length, and the fact that the equations are valid only for channels or pipes the inlet and outlet chambers are excluded and the model is approximated as the channels. These values are valid for both concept 3 and concept 5.

To calculate the heat transfer coefficient, the formula for the Nusselt number from (Storck, Karlsson, Andersson, Renner, & Loyd, 2012), is used.

$$Nu = \frac{hL}{k}$$

Formula 1

Which can be rewritten as

$$h = \frac{Nuk}{L}$$

Formula 2

to determine the heat transfer coefficient from the Nusselt number.

The Nusselt number is determined from the formula for turbulent flow through a pipe, which is valid for Reynolds number above 2300.

$$Nu = \frac{0.038Re_d^{3/4}Pr}{1 + 1.5Re_d^{-\frac{1}{8}}(Pr - 1)}$$

Formula 3

To determine that the values for the Nusselt number is correct it was compared to the much simpler equation with lower accuracy.

$$Nu = 0.027Re_d^{4/5}Pr^{1/3}$$

Formula 4

The Reynolds number is determined from

$$Re_d = \frac{UL_c}{\upsilon}$$

Formula 5

, where U is the mean flow velocity, υ is the kinematic viscosity of the fluid and L_c is the characteristic length which, in these calculations, is d for cylindrical channels and d_h for non-cylindrical channels. d_h is the hydraulic diameter which is determined by

$$d_h = \frac{4A}{O}$$

Formula 6

Where A is the cross-sectional area of the channel and O is the circumference.

The Prandtl number is calculated from the formula

$$Pr = \frac{\mu c_p}{k}$$

Formula 7

The equations are valid for Prandtl numbers between $0.5 \leq \mathrm{Pr} \leq 2000$.

The Prandtl number can be calculated using the values in Table 10, which yields a Prandtl number of ~11,986.

The hydraulic diameter is the same as the diameter (3 mm) since the cross-section is circular.

To calculate the Reynold number the mean flow velocity, characteristic length and kinematic viscosity is needed. The mean flow velocity of the model can be found by dividing the inlet mass flow rate from Table 3 and dividing it by the number of channels which gives a mean flow velocity of ~4.40 m/s through the channels. The characteristic length is the same as the hydraulic diameter, 3 mm. And the kinematic viscosity can be found in Table 10.

This gives a Reynold number of 9 809.8, which is above 2300.

The Nusselt number can then be calculated according to Formula 3, which gives a Nusselt number of 100.82.

The heat transfer coefficient can then be calculated from Formula 2 as 13 940.

The fluid properties used for the heat transfer analysis is based on the inlet temperature, normally it would be based on the mean temperature of the fluid but since the only known temperature is the inlet temperature and the objective is to determine the outlet temperature this doesn’t work. The properties would change with temperature and affect the heat transfer in the fluid throughout the cooler. The effect of the temperature on the heat transfer is therefore investigated.

The calculations are made for the channels between the inlet and outlet chambers since the calculations are not valid for anything except channels, the channel make up about 78-85% of the total cooler length and will responsible for most of the heat transfer.

First a sensitivity analysis was made on the heat transfer coefficient to determine which parameters have the greatest affect, see Figure 5.

	D	L	u	cp	kf	uu	v
h1	-3.1681e+05	3.3267e+05	-237.33	-102.82	-1536.2	-2.5568e+05	7.063e+08
h2	-2.8209e+05	3.3267e+05	-211.32	-79.902	-1730.5	-1.9868e+05	6.2889e+08
h3	-2.6637e+05	3.3267e+05	-199.55	-94.678	-1605.3	-2.3542e+05	5.9388e+08

Figure 5 - Normalized sensitivity matrix for the heat transfer coefficient

D = channel diameter
L = Characteristic length = D for cylindrical channels
u = Mean velocity of the fluid measured in practical experiments
cp = Specific heat capacity
kf = Thermal conductivity
uu = Dynamic viscosity
v = Kinematic viscosity

The parameters that have the greatest affect and are not related to the geometry of the cooler are dynamic and kinematic viscosity. Estimations based on the emitted energy from the pc box and the measured values as well as the fluid properties shows that the fluid temperature will not increase by a lot, the given condition is that the temperature should not exceed 5 degrees centigrade. The heat transfer coefficient is evaluated by changing the properties within the 5 degrees centigrade range to determine if the affect is negligible or not.

SAAB stated that the ethylene glycol concentration used was 60%, however the datasheets only include 30%, 40% and 50%. Extrapolation for 60% Dowcal100 was evaluated, see figures Figure 6-Figure 10.

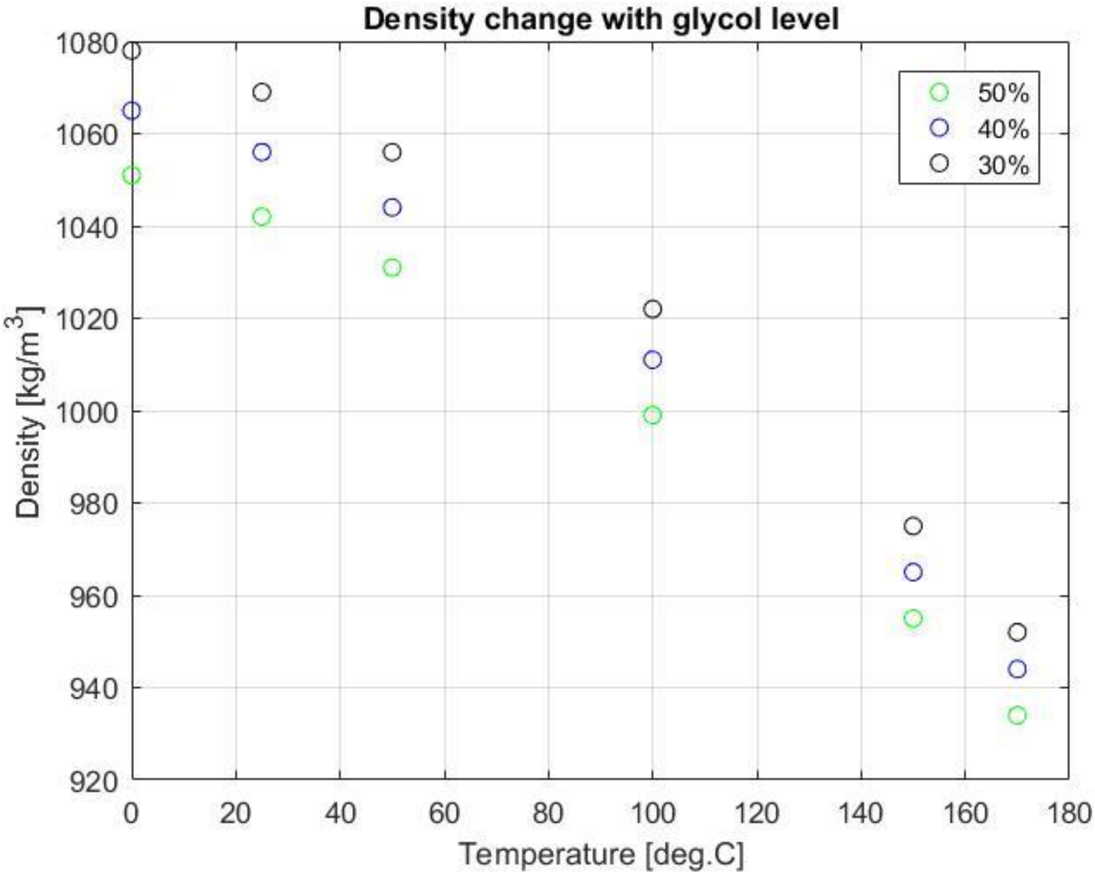

Figure 6 - Density change with ethylene glycol level

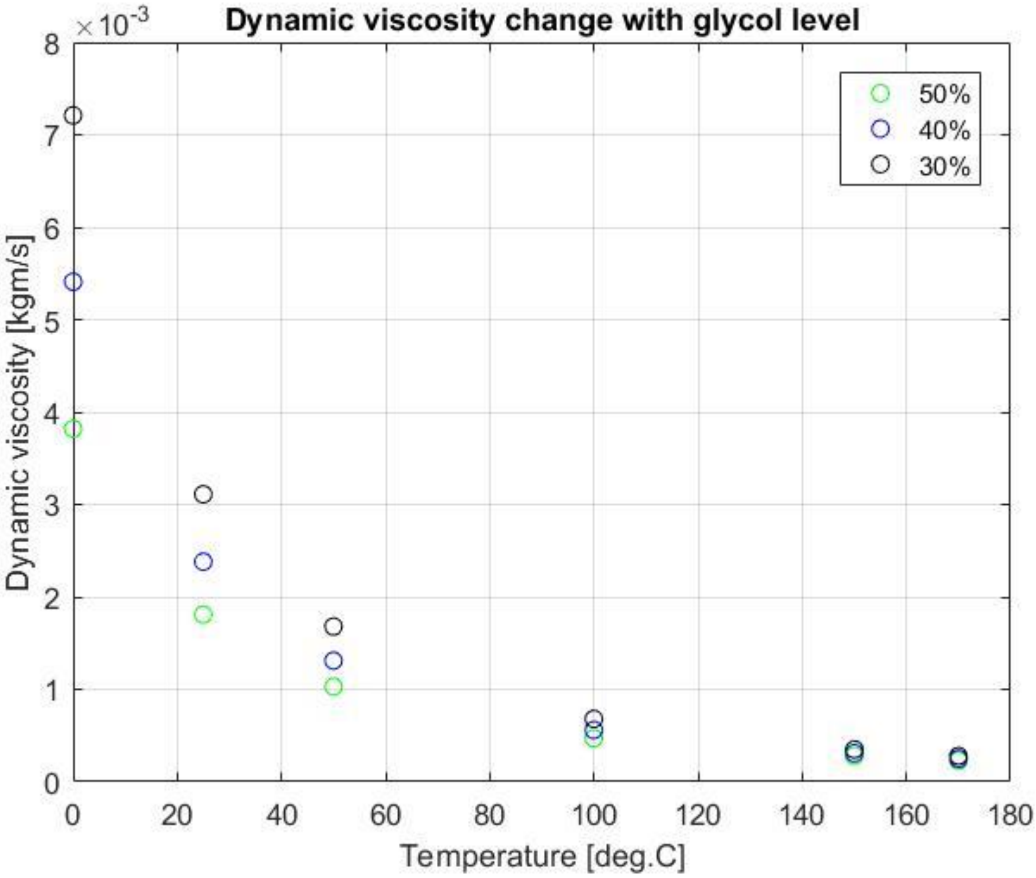

Figure 7 - Dynamic viscosity change with ethylene glycol level

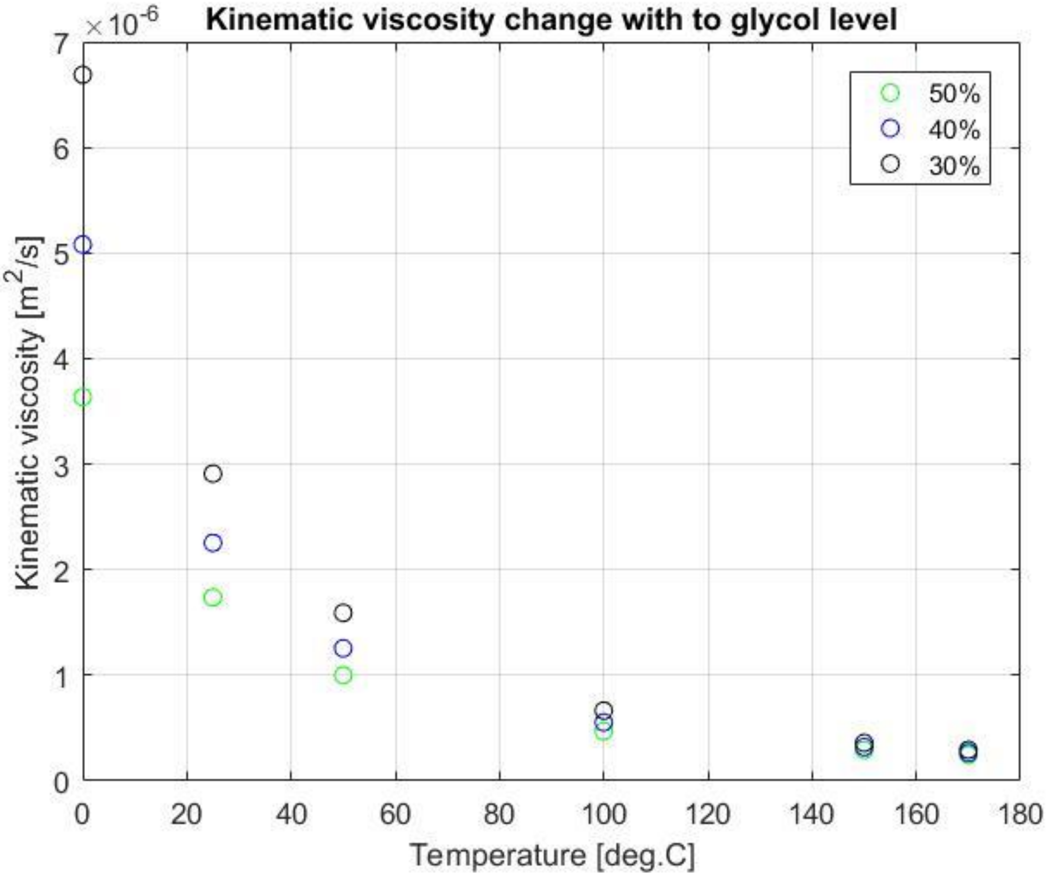

Figure 8 - Kinematic viscosity change with ethylene glycol level

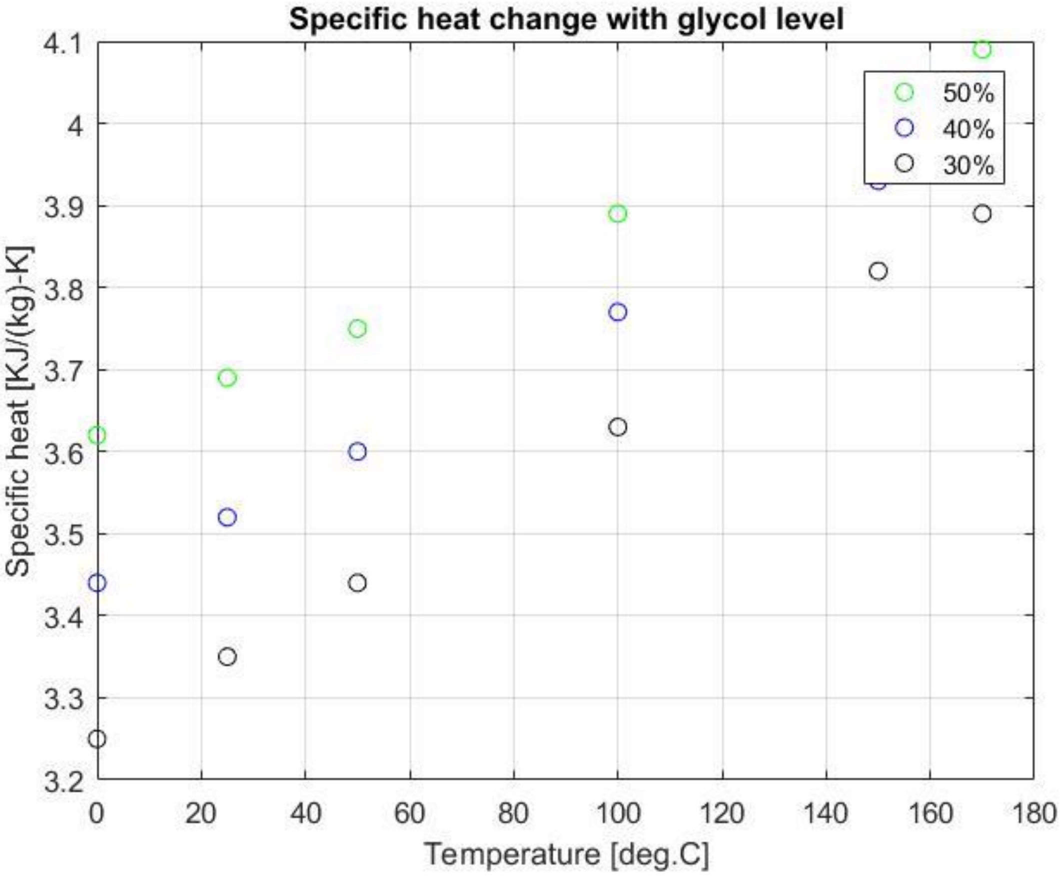

Figure 9 - Specific heat change with ethylene glycol level

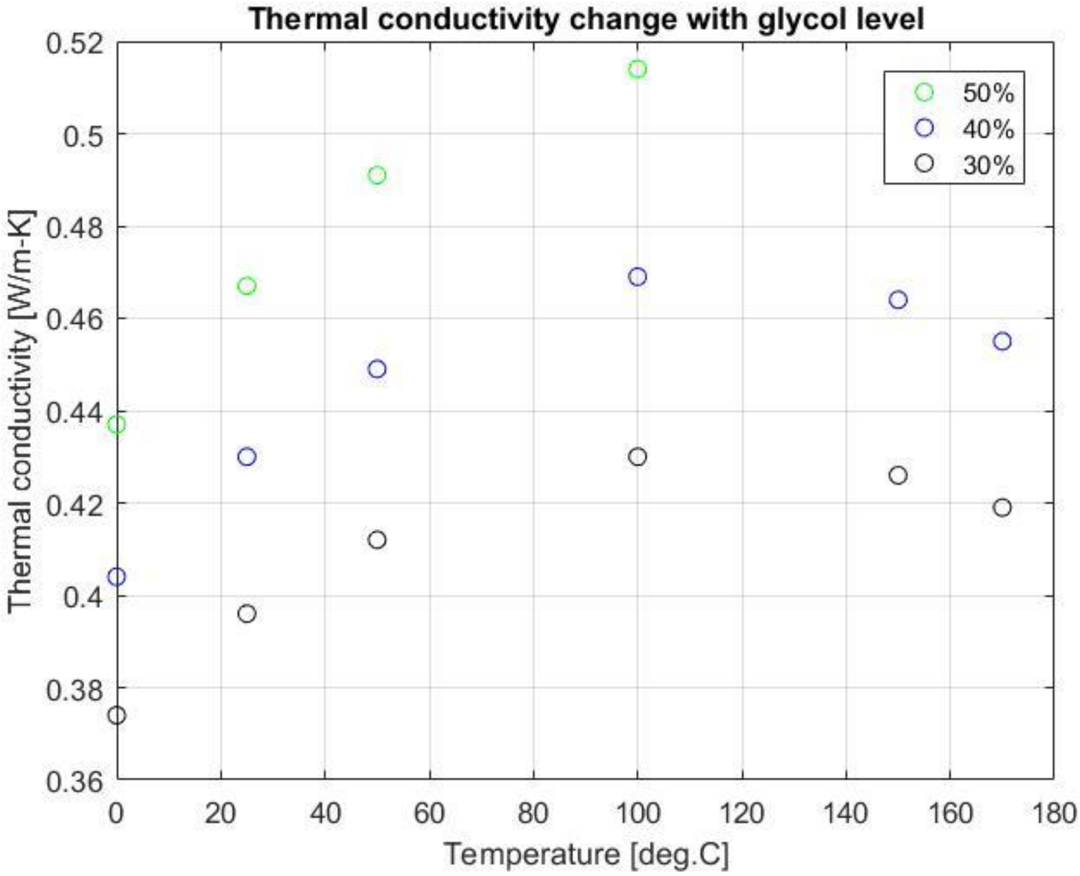

Figure 10 - Thermal conductivity change with ethylene glycol level

The ethylene glycol concentration has considerable effect on the parameters, even greater than temperature change. Since the change in values due to ethylene glycol concentration are not linear and the amount of data points to extrapolate from are few it is deemed to be a bad idea. The analyses will therefore utilize 50% ethylene glycol concentration but can be easily changed once the correct parameters are known.

To validate that the heat transfer is not considerably affected by the temperature interval, the change in the heat transfer coefficient is calculated for parameter values in the temperature interval of 70-75 degrees centigrade. Since the data only provides values at specific temperatures an interpolation needs to be made to get the values at 70- and 75 degrees centigrade. Polynomial curve fitting was evaluated and discarded since the amount of data points made it impossible, see Figure 11.

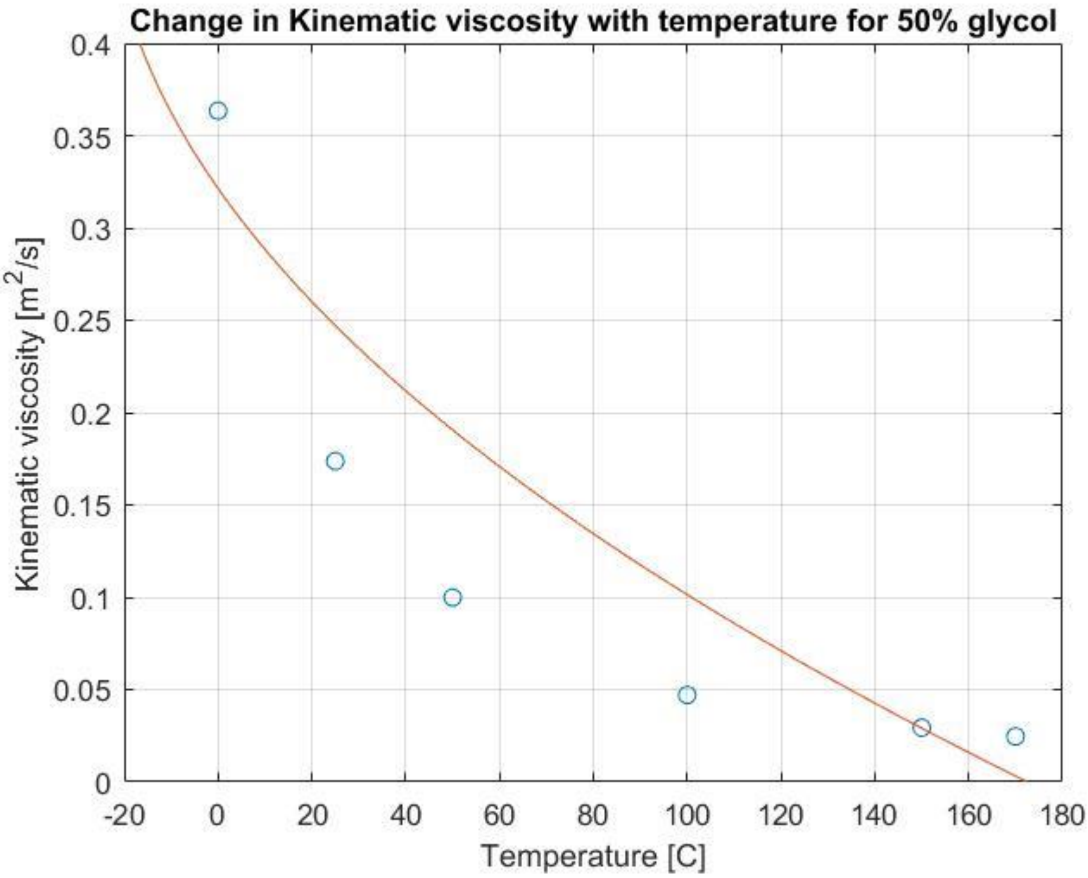

Figure 11 - Polynomial curve fitting of data points of parameters for 50% ethylene glycol concentration

Figure 8 shows that the parameter does not change linearly with temperature, however the interval used for interpolation (50-100) is approximated as linear in this case to be able to get the parameter value for 70- and 75 degrees centigrade. The linear approximation can be seen in Figure 12.

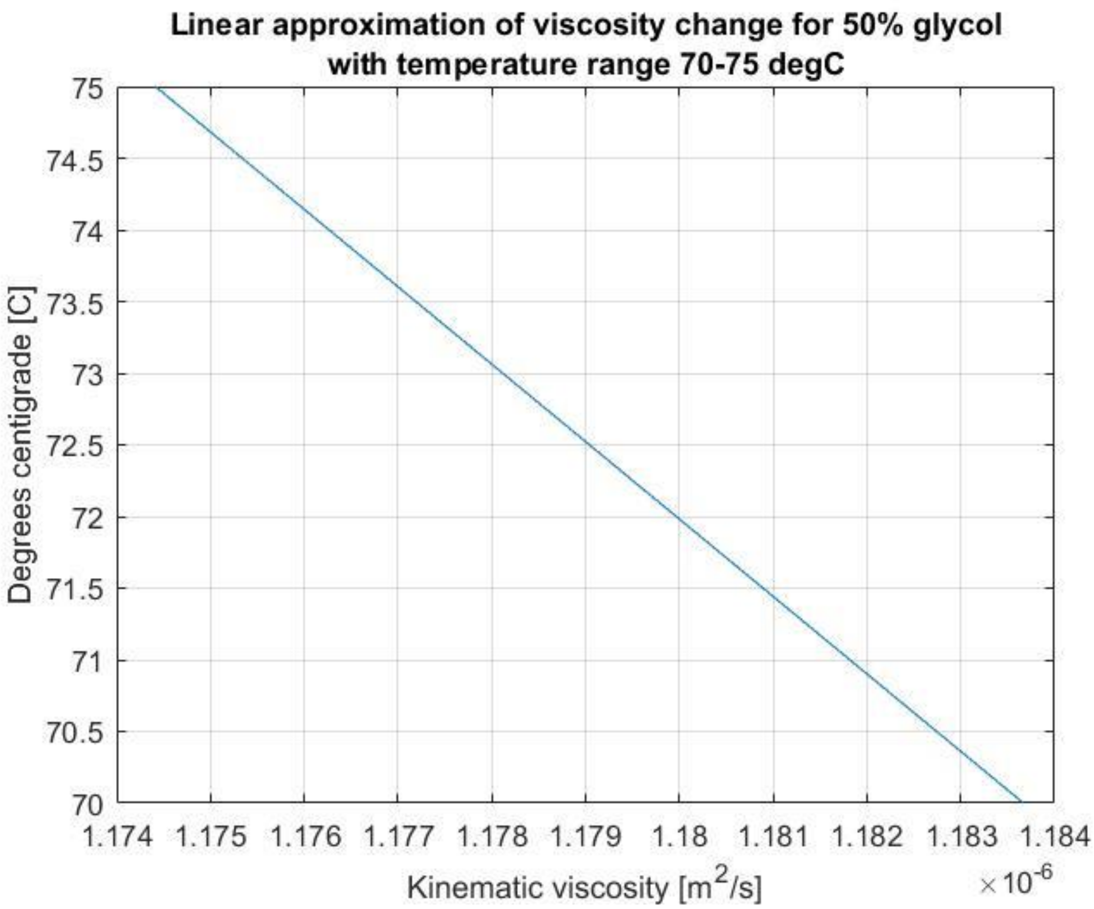

Figure 12 - Value change of kinematic viscosity in linearly approximated temperature interval

The kinematic viscosity has the greatest effect on the heat transfer coefficient according to the sensitivity analysis in Figure 5. The kinematic viscosity does not change a lot in the temperature interval but the effect on the heat transfer coefficient can not be assumed to be negligible. Therefore the heat transfer coefficient needs to be evaluated in this interval.

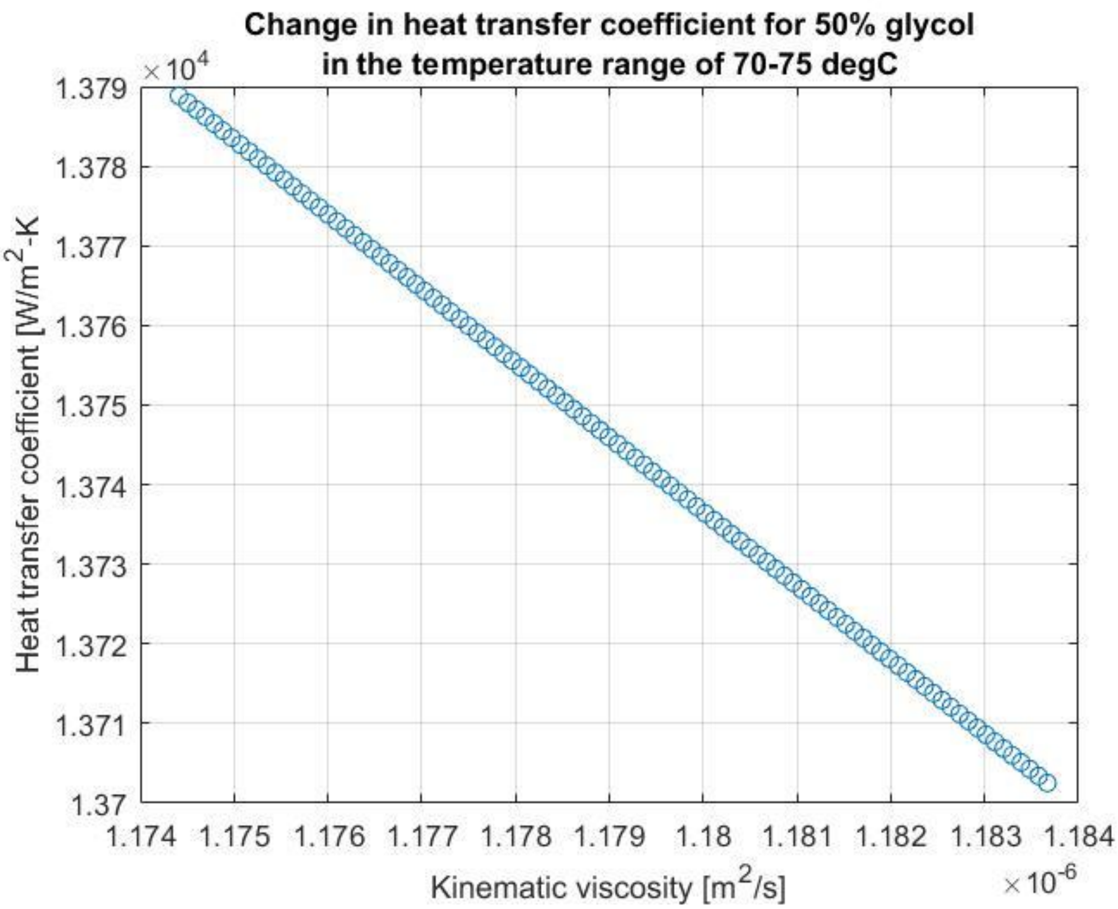

Figure 13 - Change in heat transfer coefficient in the linearly approximated temperature interval

Figure 13 tells us that the heat transfer coefficient increases with 90 W/m^2-K, or approximately 0.65% of the total value. Which means that the bulk mean temperature of the fluid can be approximated as the inlet temperature of 70 degrees centigrade for Dowcal100 at 50% ethylene glycol concentration.

3.2. Concept 3

The design parameters for concept 3 that was found during the design exploration are as follows; 31 channels, 16 on the above layer and 15 on the lower, 4,1 mm channel diameter and 2 mm wall thickness. The changeable parameters for the models can be found in the main report, under chapter 4.2.1 – Computer Aided Design.

3.2.1. One channel simplification

The entire model was simplified as channels for the design exploration, this meant that the heat was distributed evenly across all the channel surface. The Ansys simulation was done for one channel and the mass flow was divided evenly across all channels. The idea is to evaluate the best solution assuming optimal conditions, i.e. that the flow divides evenly across all channels.

The Ansys framework uses standard tetrahedral mesh with an element size range of 0.01 – 1 mm, it has a mesh sweep setting with quad mesh to get finer mesh closer to the surface and also has inflation. The mesh can be seen in Figure 14.

Figure 14 - Concept 3 one channel mesh

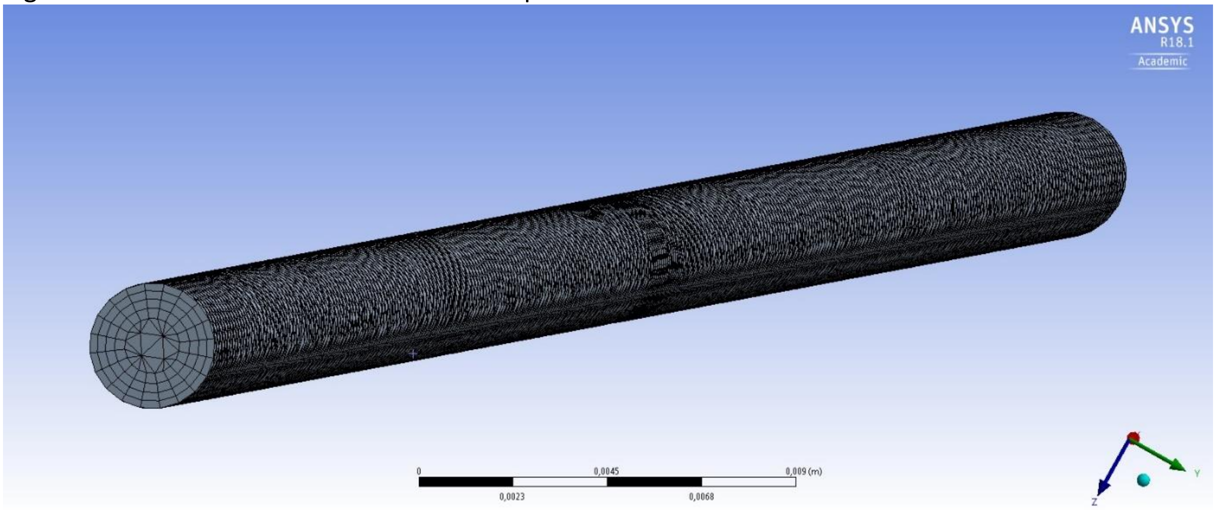

Figure 14 - Concept 3 one channel mesh

The CFX fluid model uses heat transfer with thermal energy with k-Epsilon as turbulence option.

3.2.2. Full model analysis

The full model has a mesh size of 1 mm since a finer mesh seemed to break the analysis, it also uses an inflation. The CFX fluid model uses thermal energy as heat transfer option with k-Epsilon as the turbulence option. The mesh can be seen in Figure 15 - Concept 3 full model meshFigure 15.

Due to the inlet, outlet and channels being tangent to the solid surfaces the mesh will not work with the original CAD design, therefore the inlet and outlet channel diameter is reduced by 2 mm and the channels are given an equation driven offset from the solid walls which offsets the channels by half the diameter.

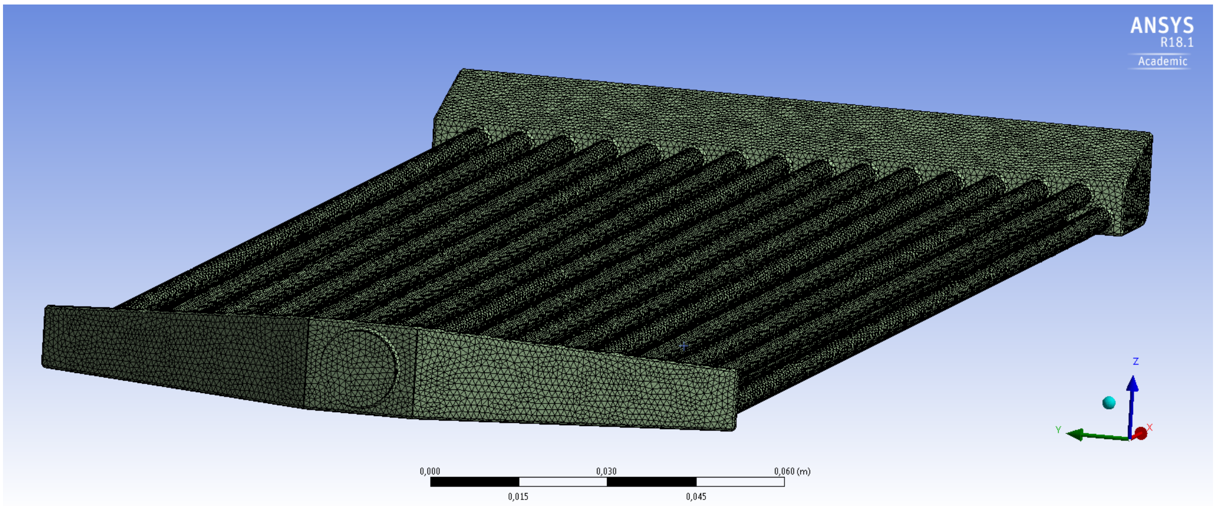

Figure 15 - Concept 3 full model mesh

3.2.3. Results

The results from the design exploration gave the following parameters for the Ansys simulation and CAD model:

Table 11 - Concept 3 Ansys input parameters

Parameter Name	**Value**	**Unit**
Density	1039.8	kg/m^3
Heat Flux	4553.5	W/m^2
Molar Mass	44.122	kg/kmol
Specific Heat	3,516	kJ/kg-K
Thermal Conductivity	0.4148	W/m-K
Dynamic Viscosity	0.001414	kg/ms
Inlet Temperature	343.15	K
Mass Flow Rate	0.9371	kg/s

Table 12 - Concept 3 CAD input parameters

Parameter Name	**Value**	**Unit**
Nr_of_layers_1	1	Number
Nr_of_layers_2	1	Number
Nr_of_pipes_Hor	16	Number
Pipe_Diameter	4.1	mm
Thickness	2	Mm

For the one channel approximation the mass flow rate is divided evenly across all channels which for this design is 31 channels. Meaning the mass flow rate through the one channel is 0.03023 kg/s.

Concept 3 has 31 channels with a diameter of 4.1 mm (which can be seen in Table 12), this gives a temperature increase of approximately 0.3 degrees centigrade and a mean flow velocity through the channels of about 4 m/s.

The full model analysis has a temperature increase of approximately 0.42 degrees centigrade and a mean flow velocity in the range 1-4 m/s. Which can be compared to the theoretical mean flow velocity which is 2 m/s assuming equal flow in all channels.

Note that the highest temperature recorded in the legend in Figure 16 are on the channel surfaces and does not represent the fluid bulk mean temperature at the outlet. In Figure 18 the legend also displays the highest recorded temperature which is in a hotspot around the turbulent areas (see Figure 20and Figure 21) and does not represent the fluid bulk mean temperature at the outlet.

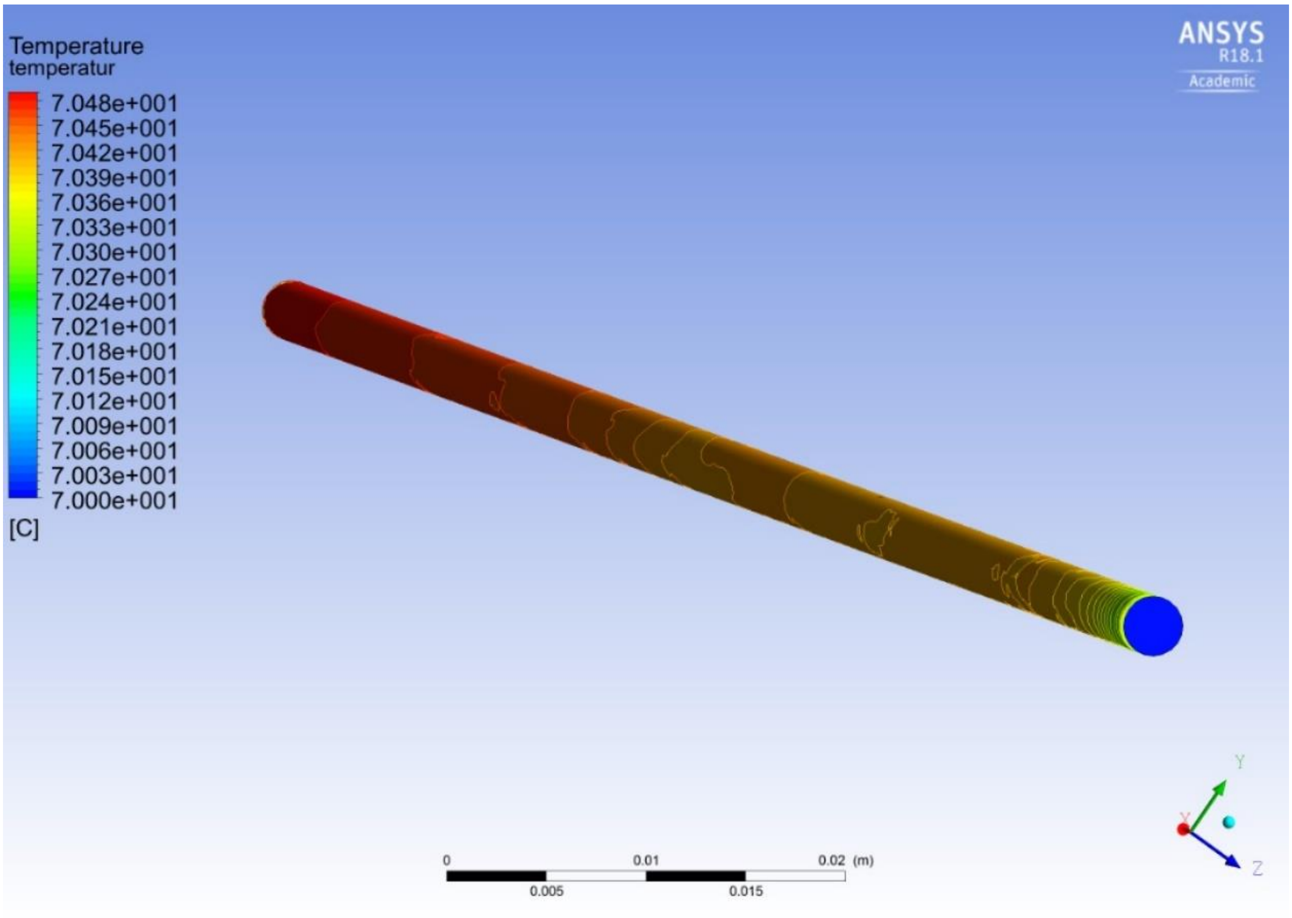

Figure 16 - Concept 3 one channel temperature results

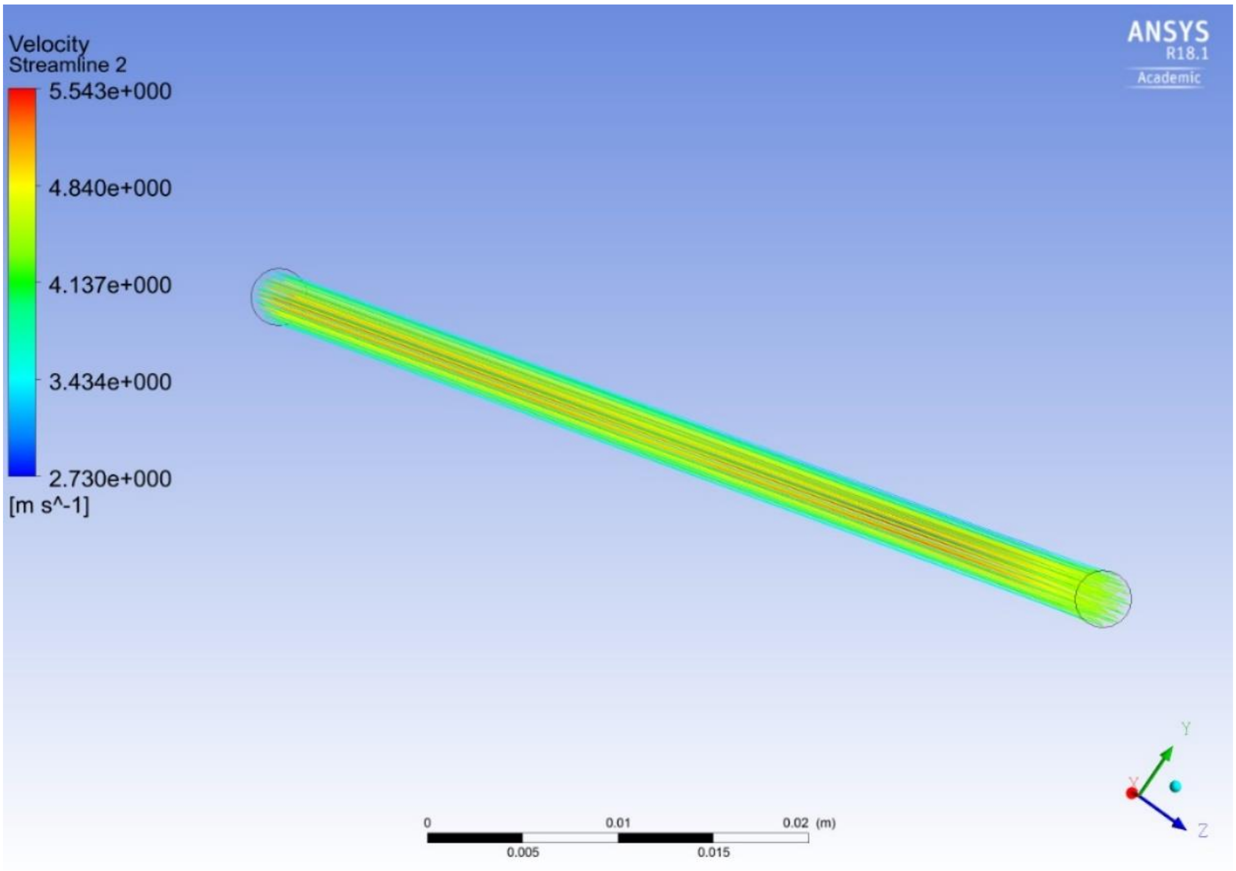

Figure 17 - Concept 3 one channel flow results

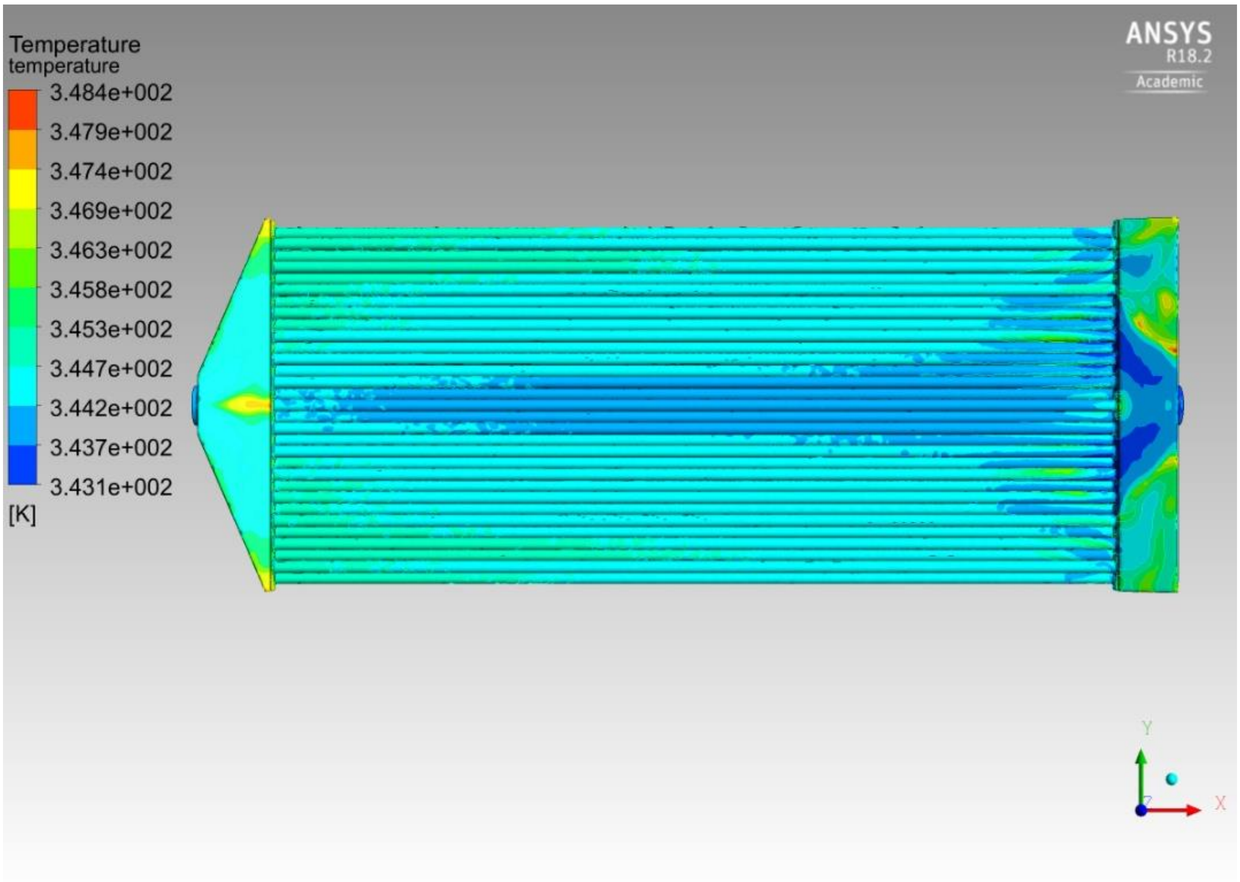

Figure 18 – Concept 3 temperature variation in fluid

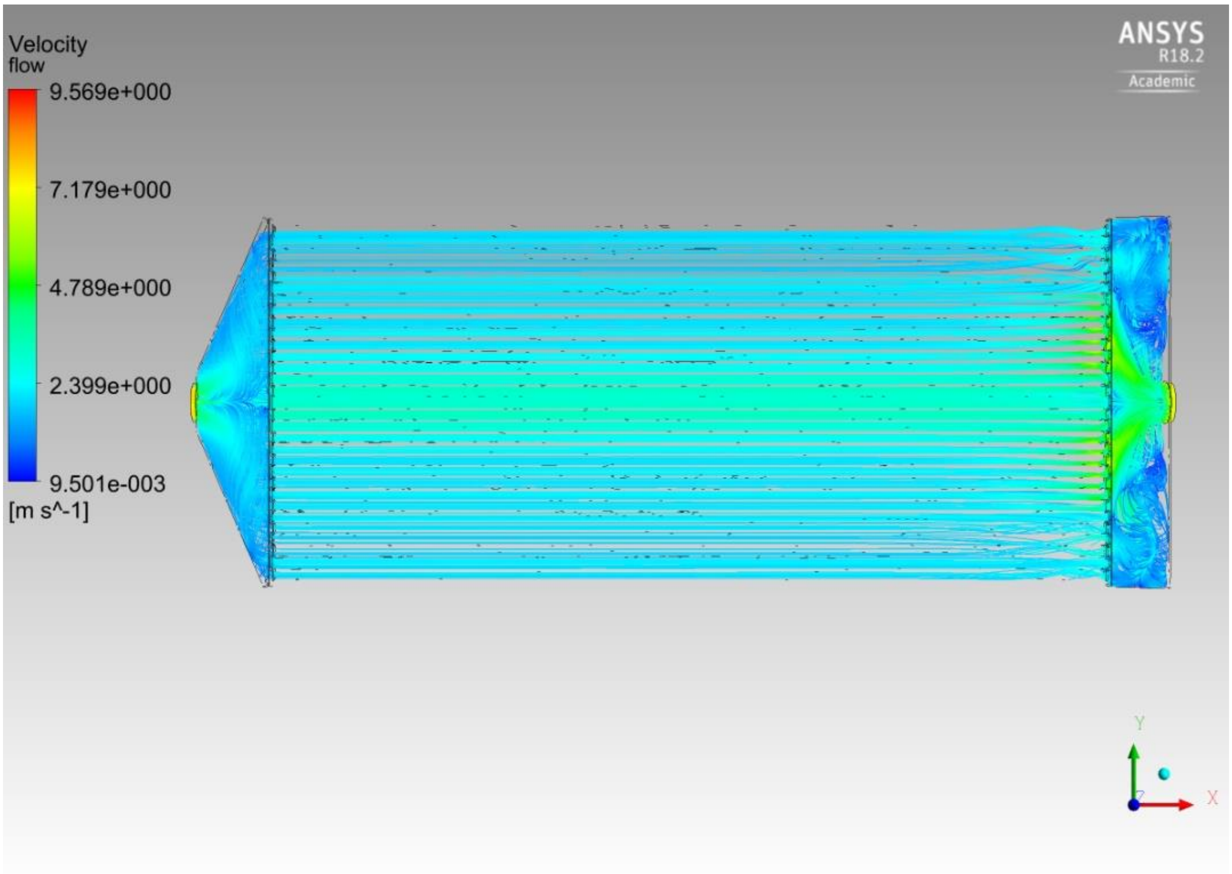

Figure 19 – Concept 3 flow velocity

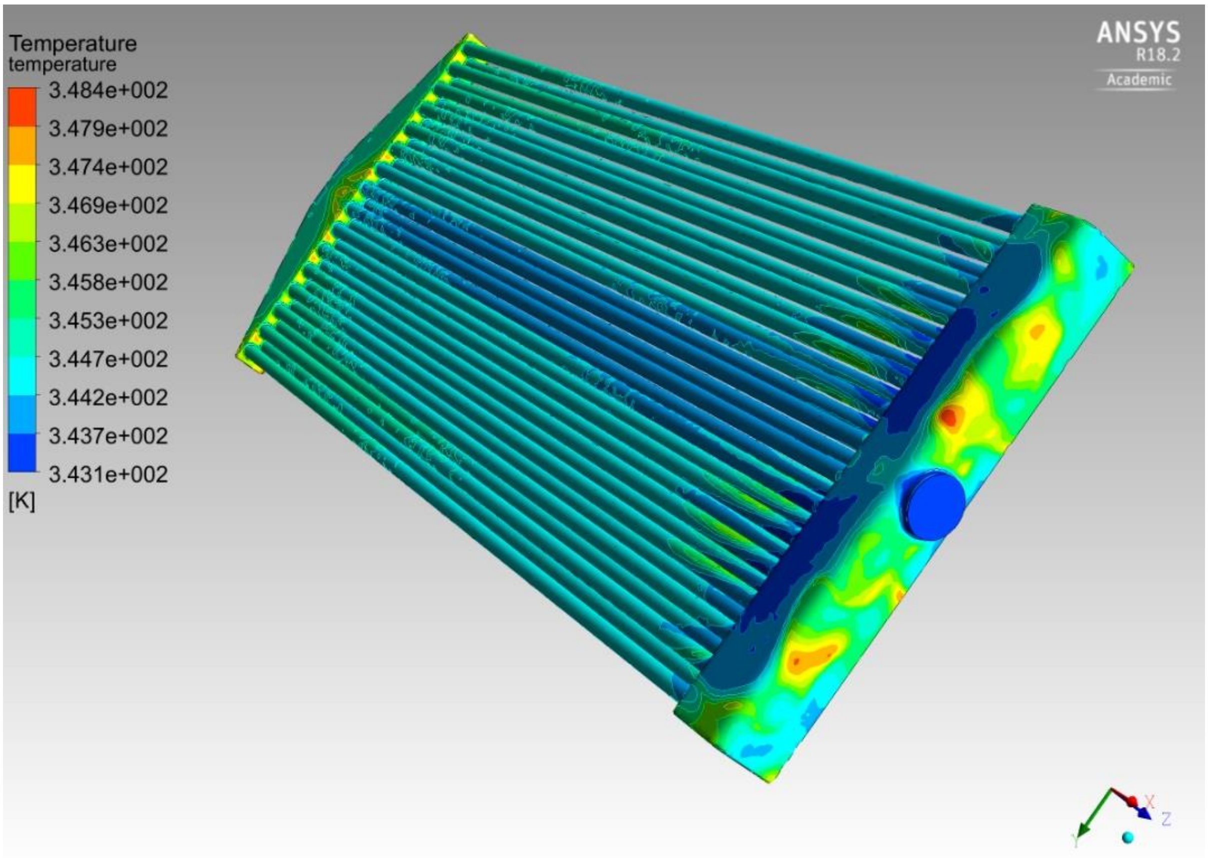

Figure 20 – Concept 3 hot spots in fluid due to flow circulation

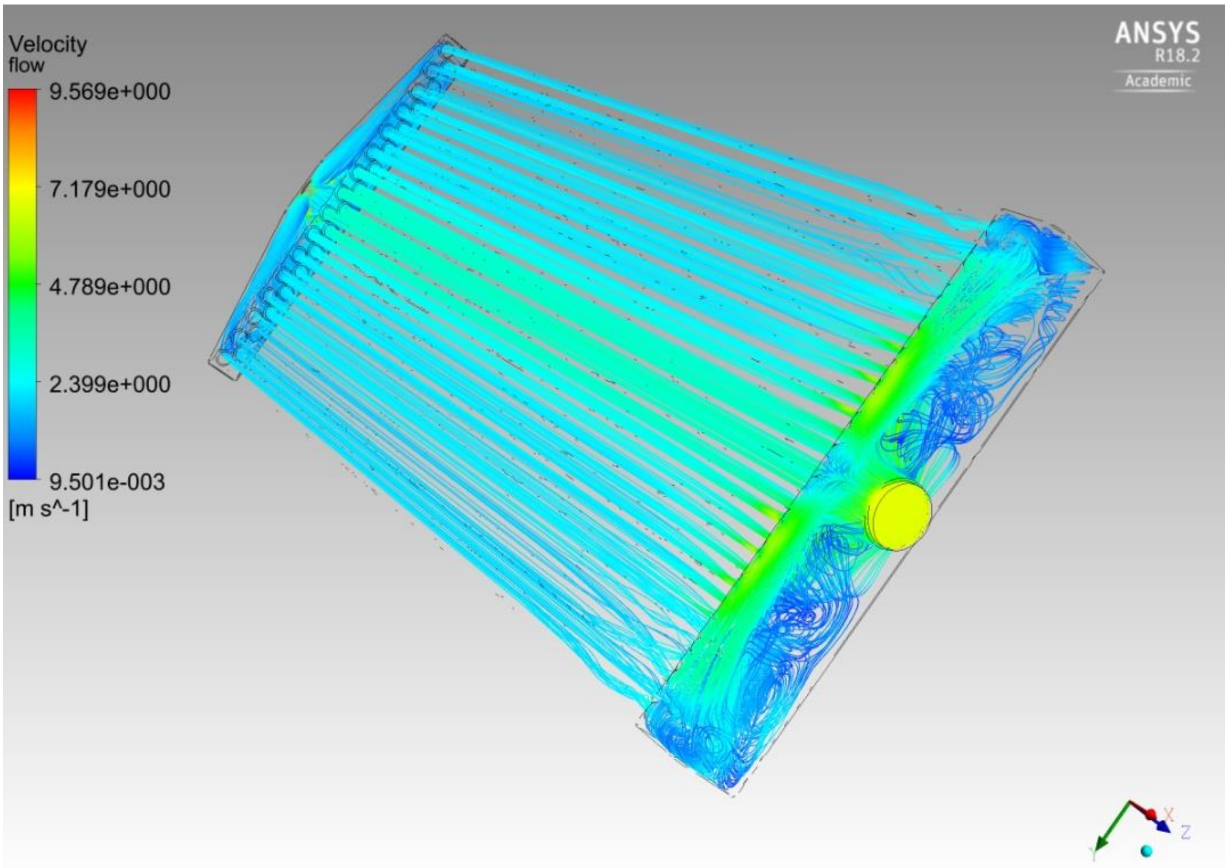

Figure 21 – Concept 3 flow velocity around hot spots at the inlet

3.3. Concept 5

The design parameters for concept 5 that was found during the design exploration are as follows; 6 channels, 13.1 mm wide and 5 mm in height with 2 mm wall thickness. The changeable parameters for the models can be found in the main report, under chapter 4.2.1 – Computer Aided Design.

3.3.1. One channel simplification

Same as with concept 3, this concept was also simplified as one channel for the design exploration. However, this concept has the drawback of not having identical channels which will be discussed later.

This analysis uses the software Ansys CFX with quad elements with a mesh size range of 0.1 – 1 mm. The fluid model uses heat transfer with thermal energy with k-Epsilon as turbulence option. The mesh can be seen in Figure 22.

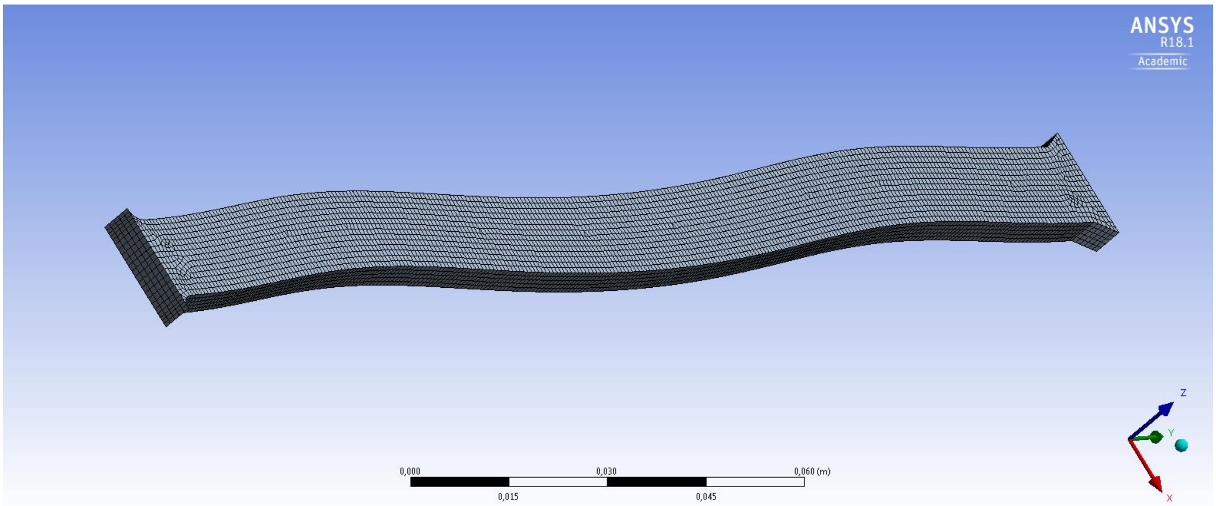

Figure 22 - Concept 5 one channel mesh

3.3.2. Full model analysis

The full model Ansys simulation uses inflation and a tetrahedral mesh with a size range of 0.1 – 10 mm. The Ansys CFX model utilizes the same settings as the one channel approximation. The mesh can be seen in Figure 23.

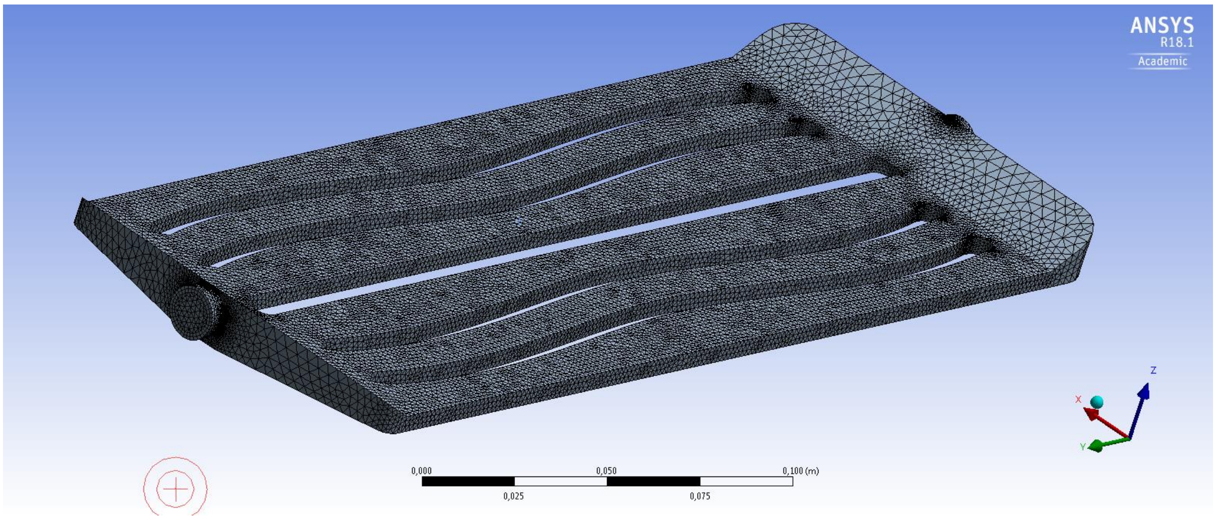

Figure 23 - Concept 5 full model mesh

3.3.3. Results

The results from the design exploration gave the following parameters for the Ansys simulation and CAD model:

Table 13 - Concept 5 Ansys input parameters

Parameter Name	Value	Unit
Density	1039.8	kg/m^3
Heat Flux	8667.9	W/m^2
Molar Mass	44.122	kg/kmol
Specific Heat	3,516	kJ/kg-K
Thermal Conductivity	0.4148	W/m-K
Dynamic Viscosity	0.001414	kg/ms
Inlet Temperature	343.15	K
Mass Flow Rate	0.9371	kg/s

Table 14 - Concept 5 CAD input parameters

Parameter Name	Value	Unit
Box_thickness	2	mm
Flange_offset_startfinish	5	mm
Flange_width	6.8	mm
Height	9	mm
Nr_of_flanges	4	Number
F_Base	13.1	mm
F_Height	5	mm
Flange_spacing	18.5	mm

For the one channel approximation the mass flow rate is divided evenly across all channels which for this design is 6 channels. Meaning the mass flow rate through the one channel is 0.1562 kg/s.

The input from Table 14 means that there are 6 channels that are approximately 13 mm wide and 5 mm high, with a channel length of 253 mm. This gave a temperature increase of 0.2 degrees centigrade and a mean flow velocity of approximately 2 m/s, the results can be viewed in Figure 24 and Figure 25.

The full model analysis has the same parameter inputs but instead have a temperature increase of 0.29 degrees centigrade and a mean flow velocity in the range of 0.1-4 m/s. The results can be viewed in Figure 26 and Figure 27.

Note that the highest temperature in the legend in Figure 25 displays the highest recorded temperature, which is at the channel surface just at the outlet and not the fluid bulk mean temperature at the outlet. The legend in Figure 26 also displays the highest recorded temperature of the fluid which is the hotpots in the turbulent areas where vortices are present, see Figure 28 and Figure 29.

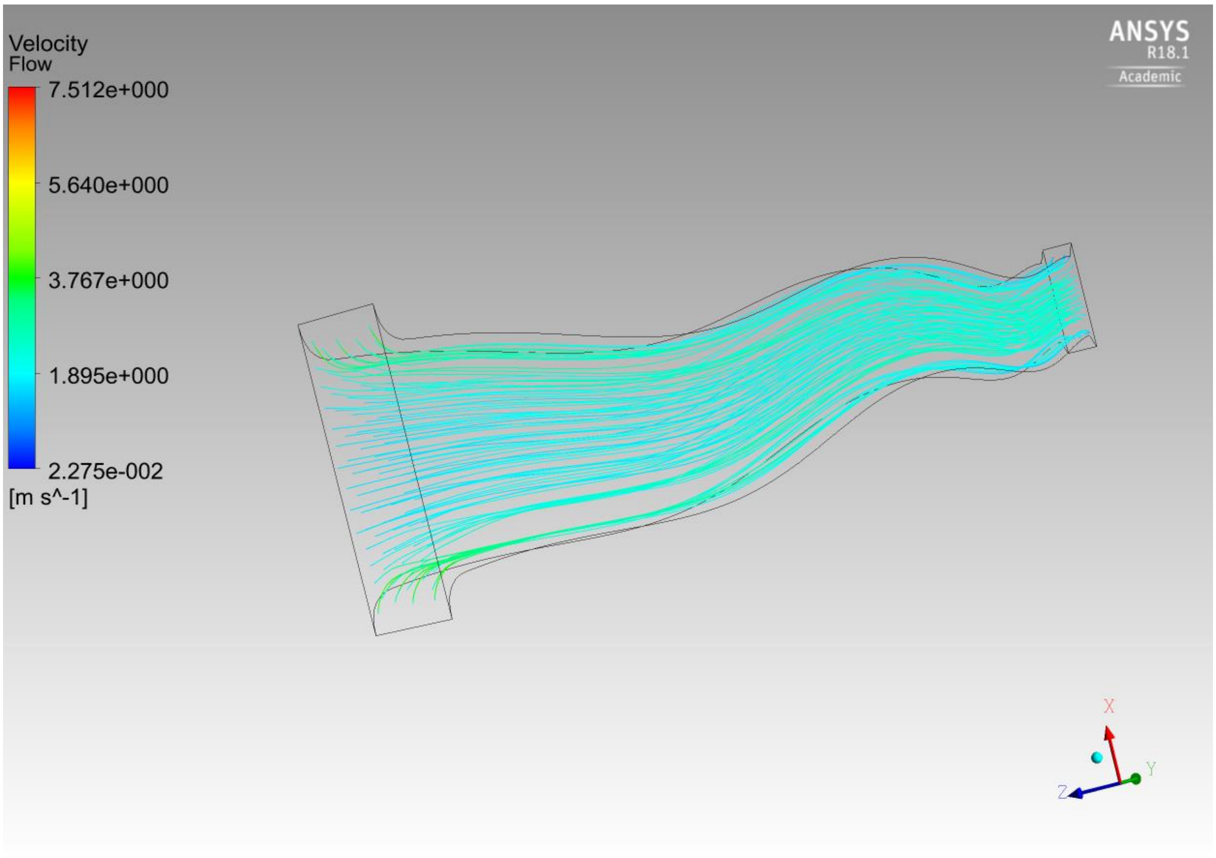

Figure 24 - Concept 5 one channel flow results

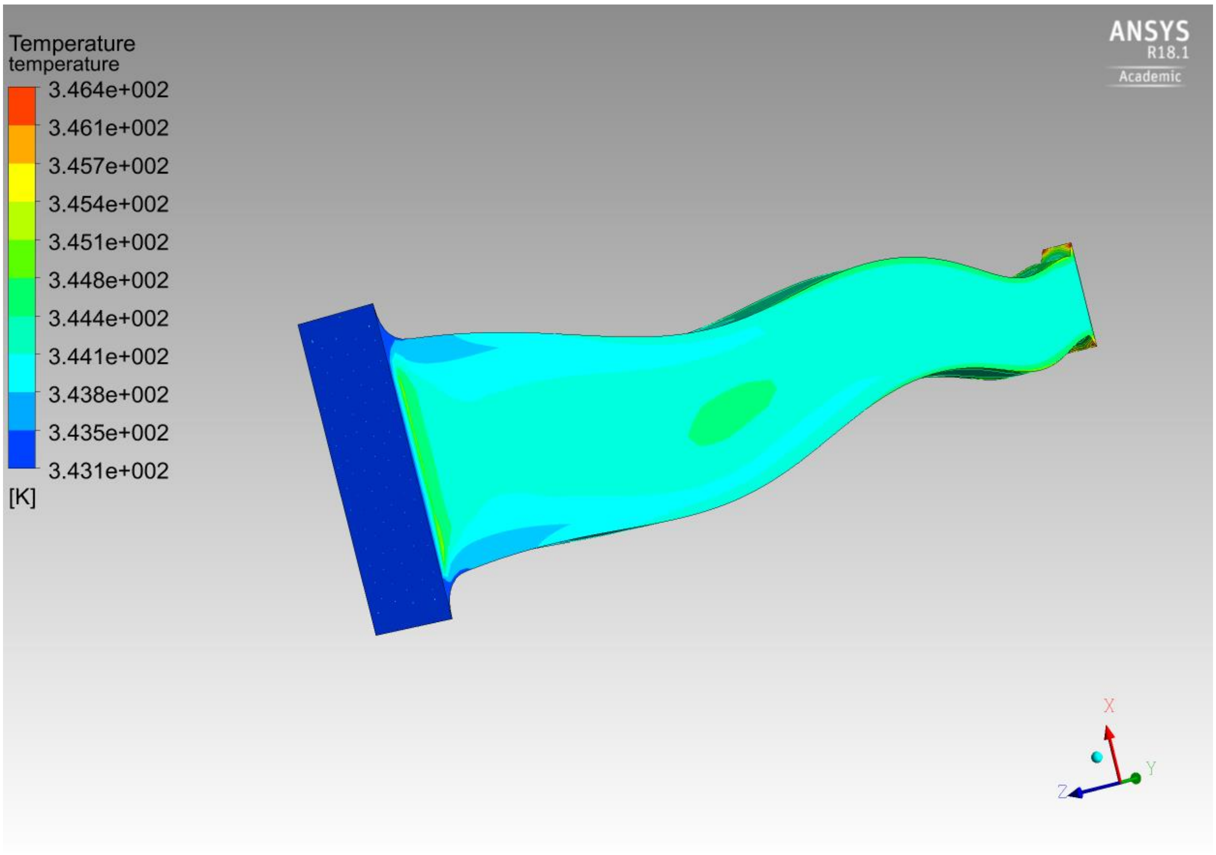

Figure 25 - Concept 5 one channel temperature results

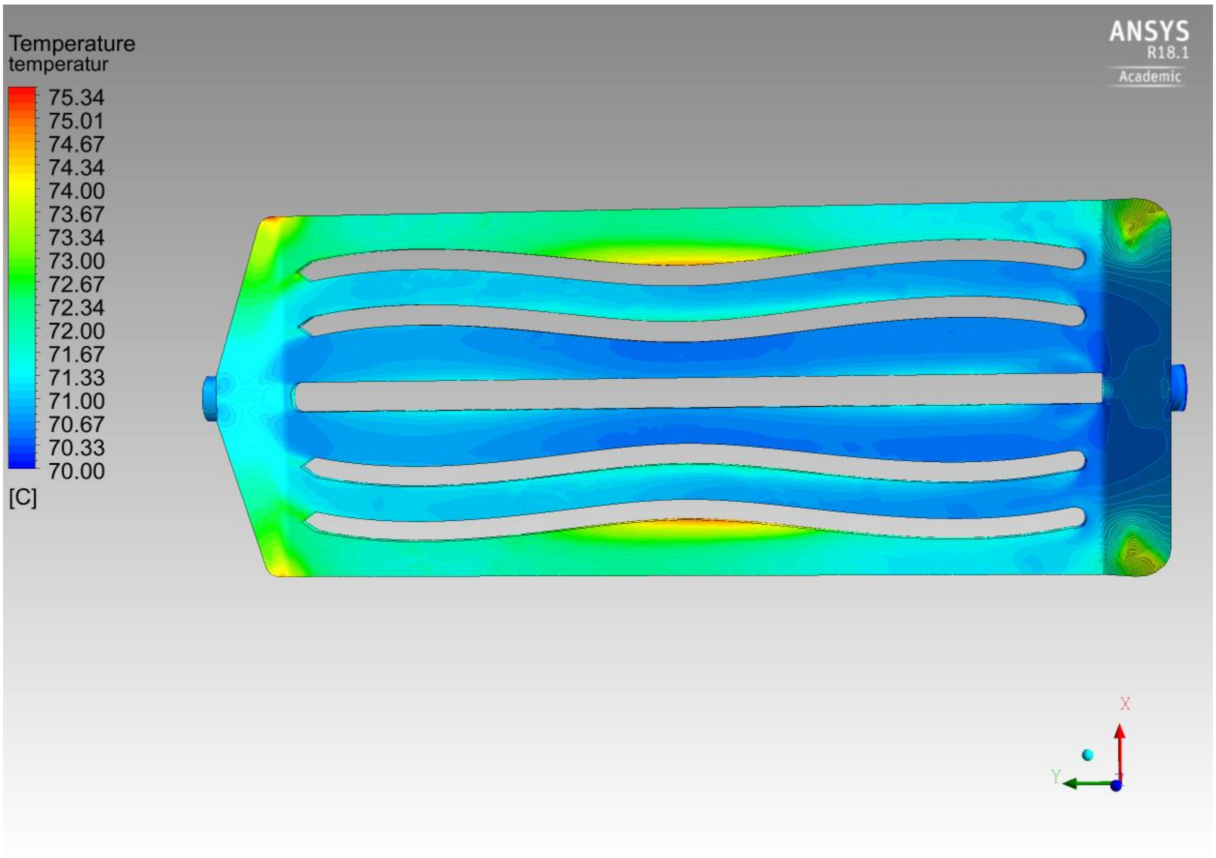

Figure 26 - Concept 5 temperature variation in fluid

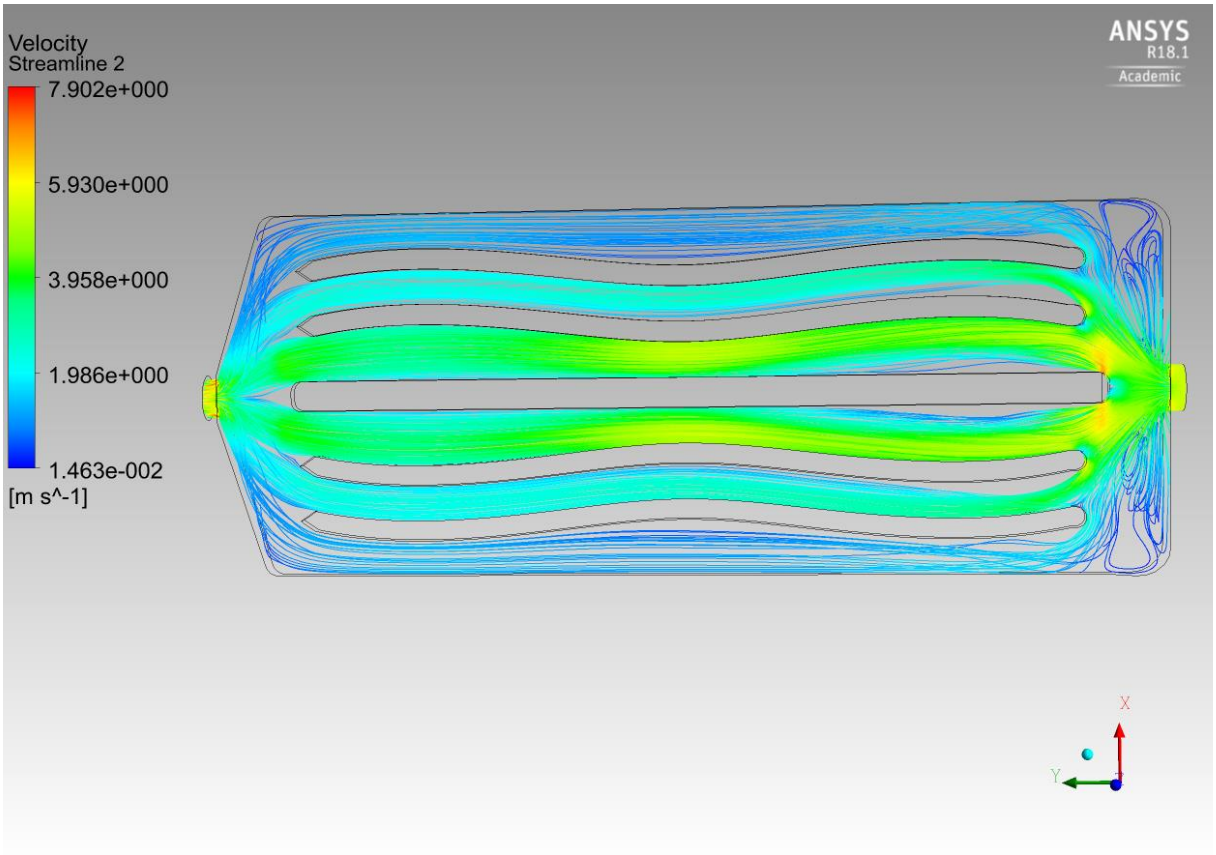

Figure 27 - Concept 5 flow velocity

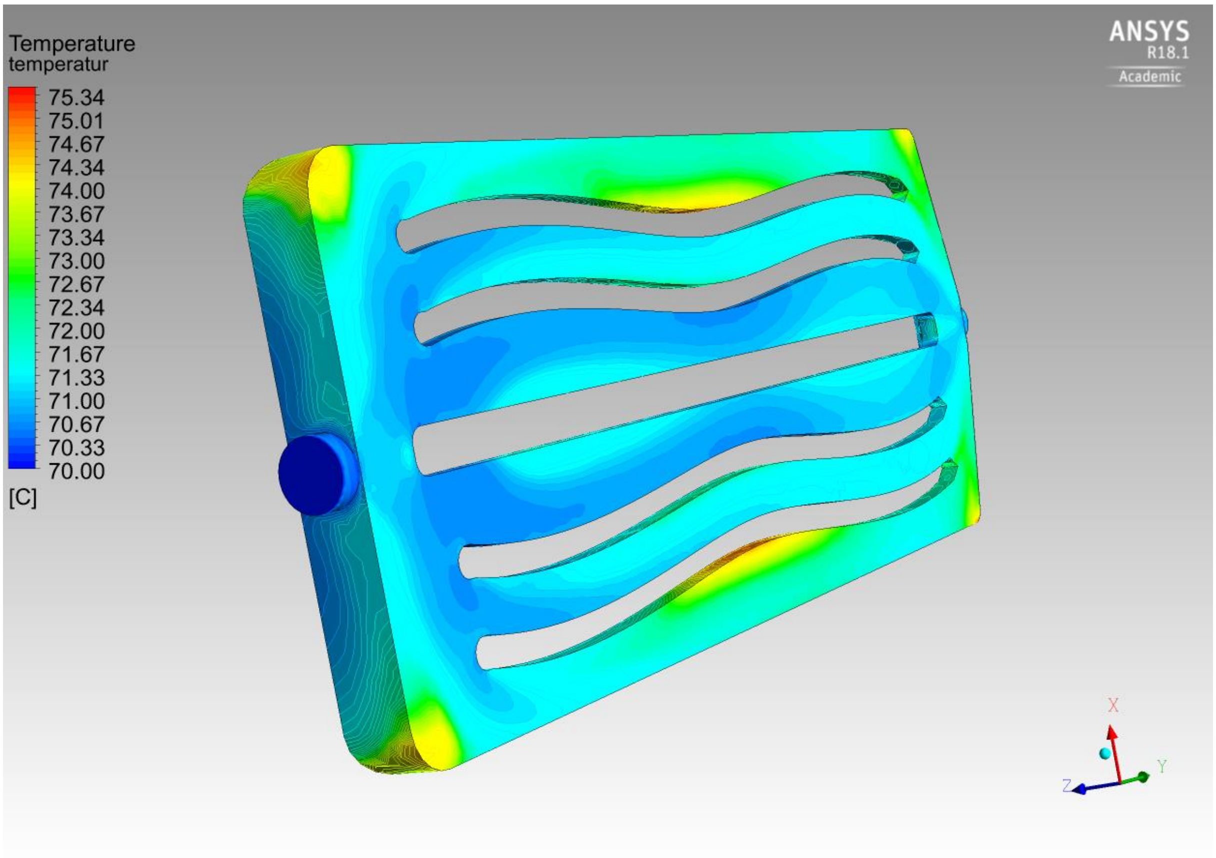

Figure 28 - Concept 5 close-up of hot spots

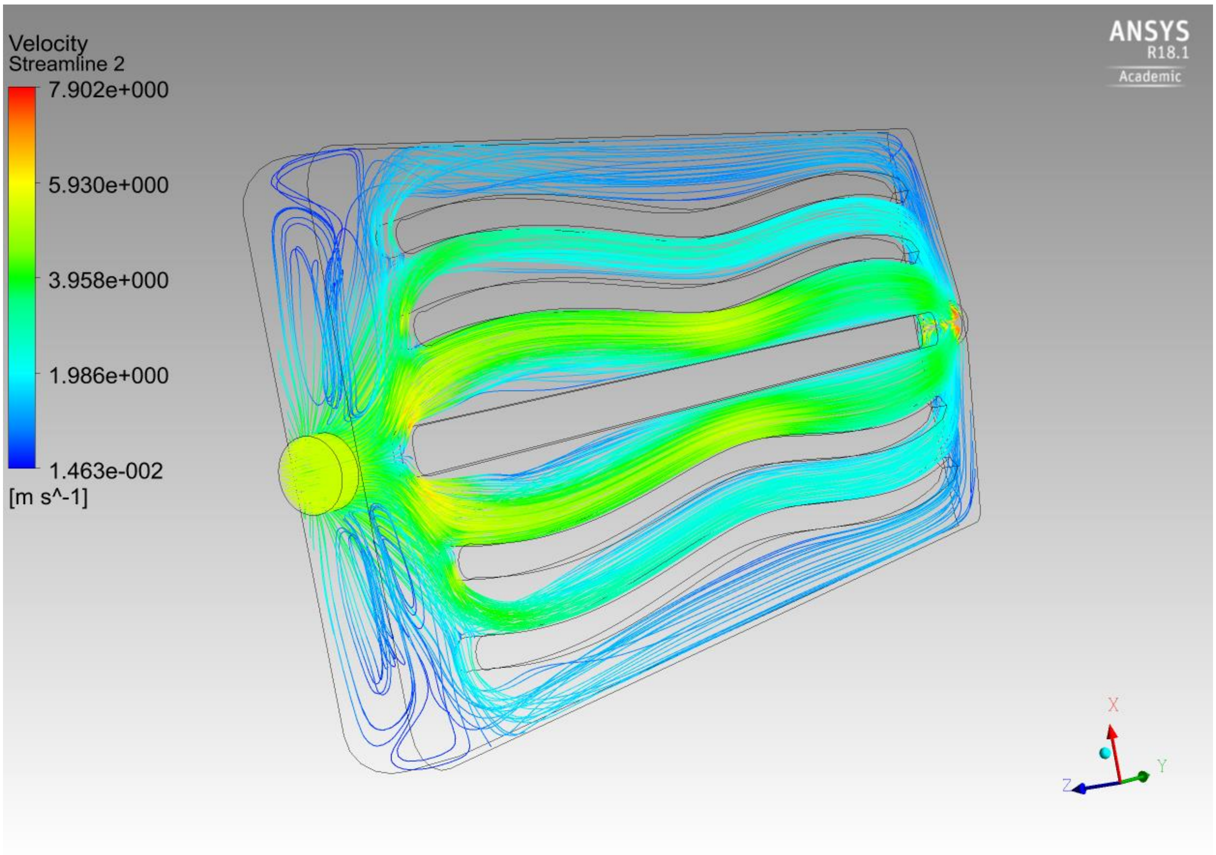

Figure 29 - Concept 5 close-up of flow around turbulent areas

4. Discussion

The error sources recorded during testing will most likely affect the results, most notably the leakage between the nippels and hose when testing the flow distribution. Which is probably the reason why the left channel showed a lower flow velocity than the right channel (see Table 5), the left channel being visibly deformed by the heat applied when connecting the hose. Test number 3 would also create a higher resistance in the channels where the outlet flow rate was measured, this can most likely be disregarded since it is only interesting to see the flow rate in the channels relative to each other. The actual numbers do not mean anything.

The effect of the leakage in the hose couplings and the deformed tubes connecting the model to the pump are deemed negligible since they will affect the outlet flows equally and only scale the values of the flow rates. However, it may affect the mass flow rate used in the simulations, but a more accurate mass flow rate can be applied easily by changing the parameter value in Ansys once it is obtained. These analyses only show the results of the given parameters which have been specified and unfortunately, they do not match the value specified by SAAB in terms of inlet pressure. Furthermore, the inlet pressure will decrease at the inlet since there is a choke to compensate for the diameter difference. The pressure is irrelevant if the flow rate at the inlet can be measured or is known.

The tests were timed manually which may have affected the results, which is why several measurements were taken and a mean calculated. The effect of this is only relevant when comparing the outlet flow rates in test number 3 since the total mass flow rate, as discussed previously, is irrelevant on its own. The tests showed fairly even results and most likely does not affect the end results in any noticeable way.

For the design exploration the flow and heat analysis were simplified as one channel where the flow was divided evenly across all channels. This gave the best theoretical cooler but necessarily not the best one in the end since flow was neglected. Had there been a criterion for minimum allowed flow velocity or max deviation from the theoretical mean flow velocity in the channels; a lot of the designs would probably have failed, among them the final design for concept 5. The issue is that when trying to minimize the weight and create a lot of channel surface area in the design exploration, it will create a higher total cross-sectional area in the channels. This means that at some point the fluid going in the manifold will continue, more or less, straight down the middle channels.

The one channel simplification for concept 5 will have slightly less accuracy than concept 3 since it does not have homogenous channels like concept 3. The effect of this simplification is very hard to determine, most likely it will create a wider gap between the results from the one channel simplification and the full model analysis. It probably has a greater effect on the fluid flow but since it is not considered in the design exploration the overall effect of this simplification is somewhat lower.

The flow distribution in the final designs are probably reasonable, comparing them to the theoretical mean flow velocity showed that it is somewhere in the middle of the flow velocity range of both concepts. The flow distribution also corresponds quite well with the practical tests done where the flow in concept 5 where considerably worse at the outer channels. Concept 3 showed a more even flow both in the simulation and the practical tests.

The idea that the lowest temperature increase is better may not be a relevant measurement since the goal specified was to not increase the fluid temperature with more than 5 degrees centigrade. There are probably better solutions in terms of flow distribution well below the 5-degree centigrade mark hidden among the vast number of DOE’s produced.

The effects of using the inlet temperature instead of the fluid bulk mean temperature have been investigated with regards to the heat transfer coefficient and deemed negligible. This is only the case for this particular heat transfer problem where it is known beforehand that the temperature increase will not exceed 5 degrees centigrade by estimations. In a more complex problem this will not work and will most likely have unpredictable and undesirable effects.

In Figure 21 and Figure 29 there are clearly visible vortices that form just after the inlet on each side. These vortices generate hotspots, these hotspots are not an issue in these results but if one were to consider the temperature of the PC box it would. If there was a limit to how high the temperature could get on the PC box at any given place these hot spots create problems. No such criterion was given and thus disregarded but will need further investigation to ensure that the cooler is able to keep the PC box at the right temperature to keep the electronics from overheating, potentially causing catastrophic failure.

5. Conclusion

Concept 5 shows a lot better cooling ability since it increases the temperature between the inlet and outlet with only 0.29 degrees centigrade. However, the flow analysis does show a problem with in the outer channels on either side where the flow velocity drops to about 0.1 m/s and creates hotspots throughout the length of the channels.

Concept 3 shows a lot better flow distribution but does increase the fluid temperature with 0.42 degrees centigrade. There are more prominent hot spots around the vortices just after the inlet which are not present in concept 5.

Both concepts are able to fulfill the goal, one better than the other but both show problems in terms of flow distribution and hot spots at certain areas.

The most critical problem areas brought up in this report are due to the simplifications and assumptions made to be able to finish, thus it is strongly recommended that the designs are further investigated by a more extensive heat transfer analysis as well as including fluid flow in the design evaluation and exploration.

References

Miller, D. (1990). *Internal flow systems.* Cranfield, Bedford: BHRA (Information Services).

Rohsenow, W. M., Hartnett, J. P., & Cho, Y. I. (1998). *Handbook of Heat Transfer.* United States of America: McGraw-Hill.

Schnabel, T., Oettel, M., & B, M. (2017). *Design for Additive Manufacturing - Guidelines and Case Studies for Metal Application.* Dresden, Germany: Fraunhofer IWU.

Storck, K., Karlsson, M., Andersson, I., Renner, J., & Loyd, D. (2012). *Formelsamling i termo- och fluiddynamik.* Linköping: LiTH.

Appendices

Appendix I – 3D printed test models

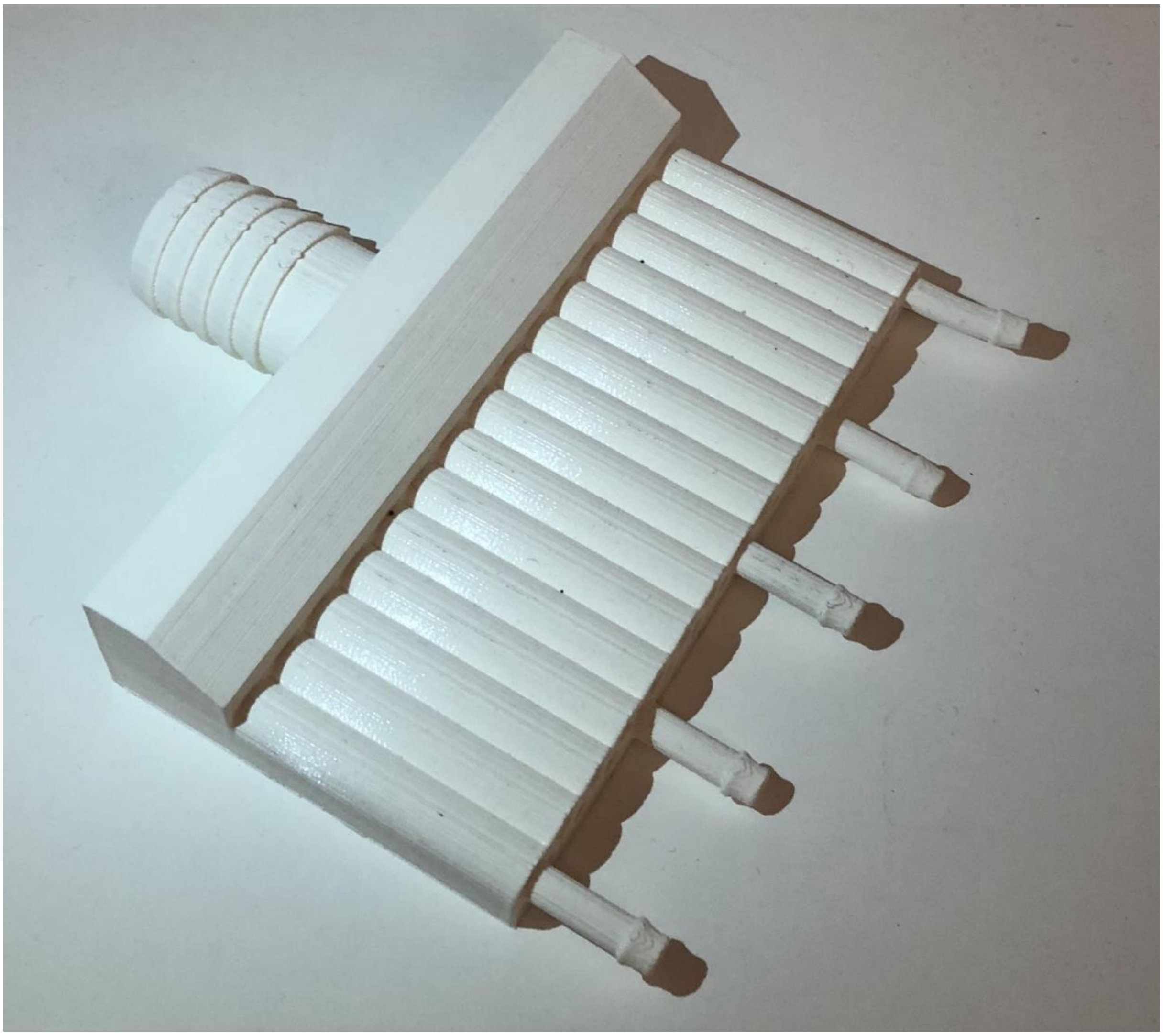

Figure 30 – Half model of concept 3 in order to measure the flow, used in test 3

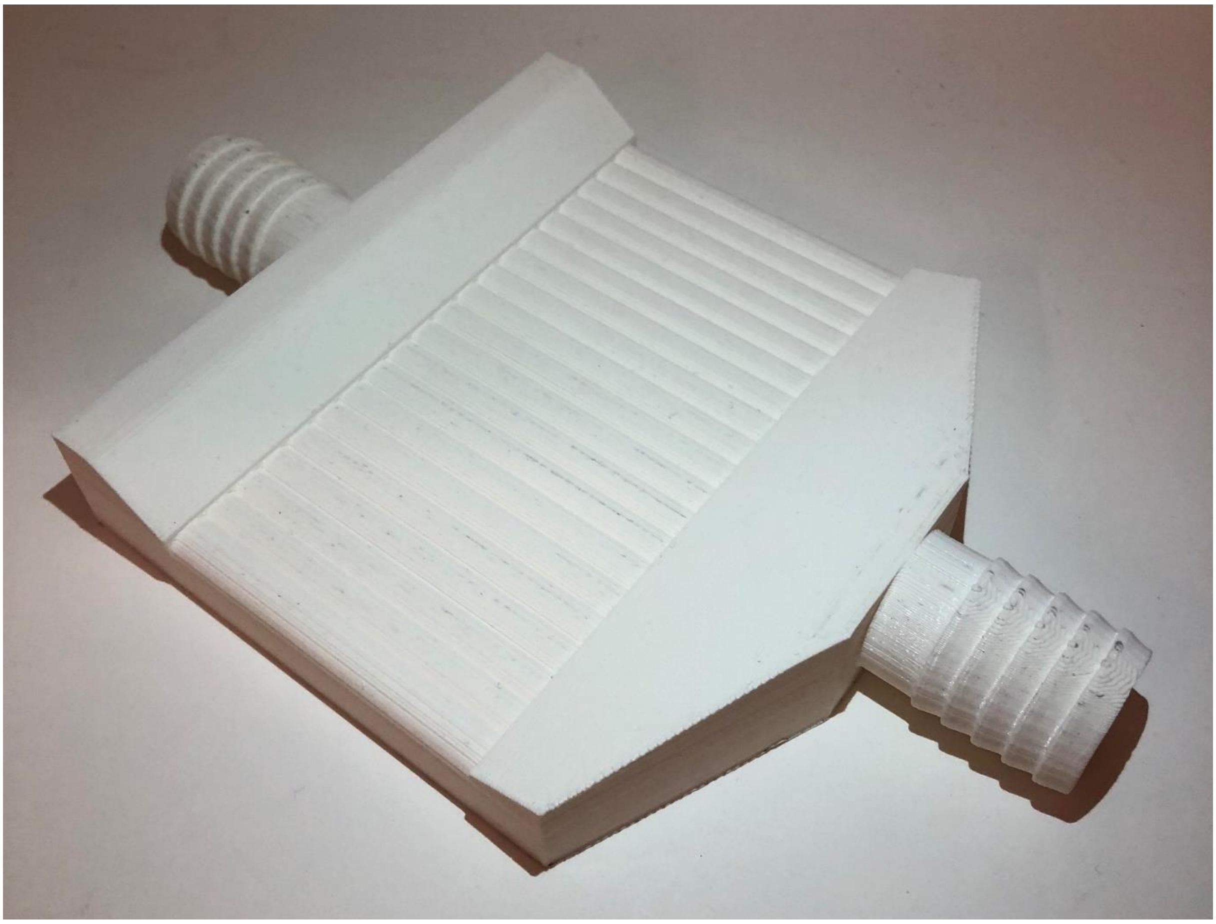

Figure 31 – Full model of concept 3, used in test 4

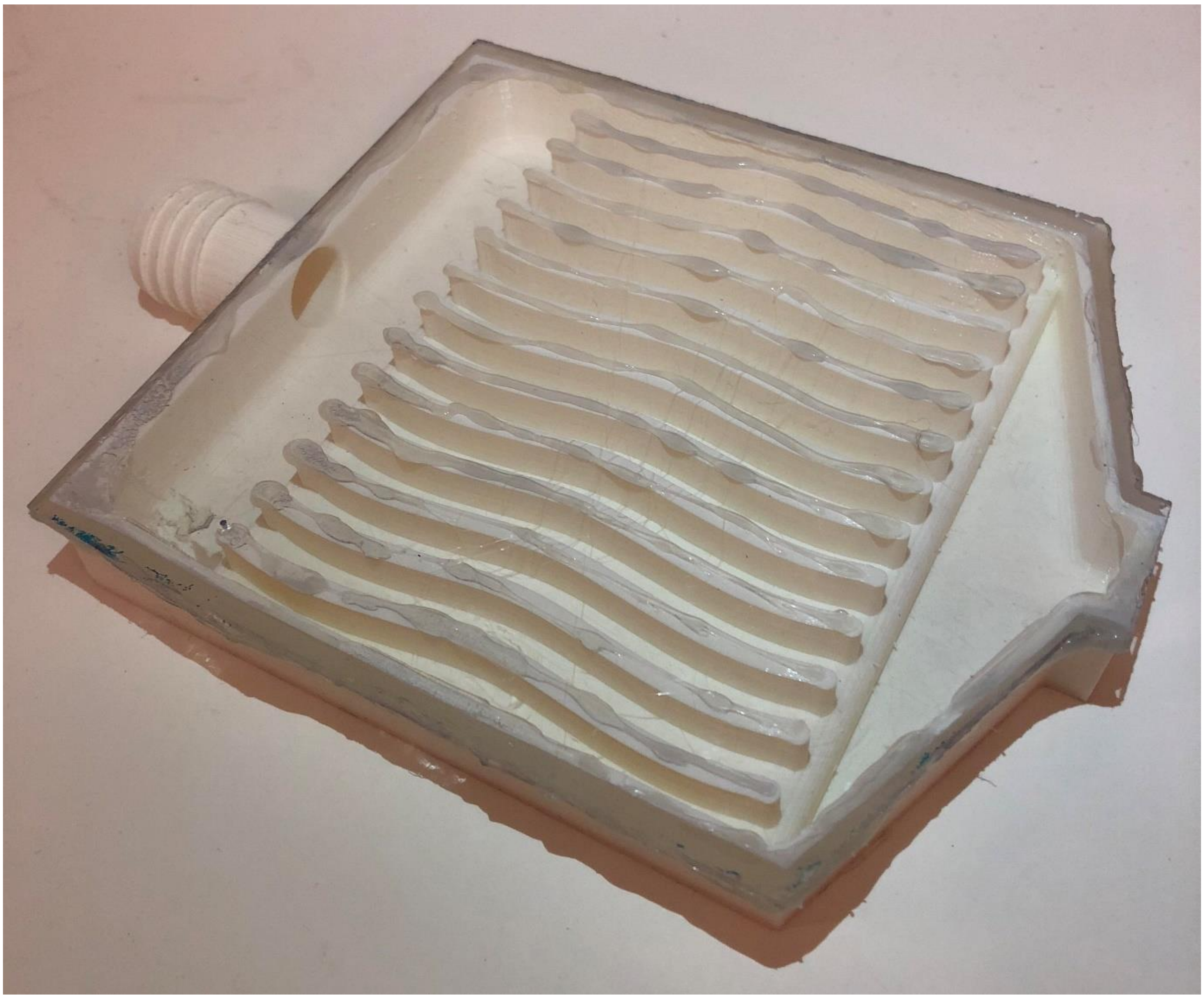

Figure 32 - Concept 5 with transparent top in order to see how the water is flowing

Appendix II – Dowcal100 property tables

DOWCAL 100 fluid at 30% Ethylene Glycol Concentration by Volume

	Temp.		Specific Heat	Density	Thermal Conductivity	Viscosity	
						Dynamic	Kinematic
	°C	K	J/(kg)(K)	kg/m^3	W/mK	[kgm/s]	[m^2/s]
1	0	273,15	3,62	1051	0,437	0,00382	3,63E-06
2	25	298,15	3,69	1042	0,467	0,00181	1,74E-06
3	50	323,15	3,75	1031	0,491	0,00103	9,99E-07
4	100	373,15	3,89	999	0,514	0,00047	4,7E-07
5	150	423,15	4,03	955	0,507	0,00028	2,93E-07
6	170	443,15	4,09	934	0,495	0,00023	2,46E-07

DOWCAL 100 fluid at 40% Ethylene Glycol Concentration by Volume

Temp.		Specific Heat	Density	Thermal Conductivity	Viscosity	
					Dynamic	Kinematic
°C	K	J/(kg)(K)	kg/m^3	W/mK	[kgm/s]	[m^2/s]
0	273,15	3,44	1065	0,404	0,00541	5,08E-06
25	298,15	3,52	1056	0,43	0,00238	2,25E-06
50	323,15	§	1044	0,449	0,00131	1,25E-06
100	373,15	3,77	1011	0,469	0,00056	5,54E-07
150	423,15	3,93	965	0,464	0,00031	3,21E-07
170	443,15	3,99	944	0,455	0,00025	2,65E-07

DOWCAL 100 fluid at 50% Ethylene Glycol Concentration by Volume

Temp.		Specific Heat	Density	Thermal Conductivity	Viscosity	
					Dynamic	Kinematic
°C	K	J/(kg)(K)	kg/m^3	W/mK	[kg/ms]	[m^2/s]
0	273,15	3,25	1078	0,374	0,00721	6,69E-06
25	298,15	3,35	1069	0,396	0,00311	2,91E-06
50	323,15	3,44	1056	0,412	0,00168	1,59E-06
100	373,15	3,63	1022	0,43	0,00068	6,65E-07
150	423,15	3,82	975	0,426	0,00035	3,59E-07
170	443,15	3,89	952	0,419	0,00028	2,94E-07